

Developing Ultra-High Performance Concrete Mix Designs For Arizona Bridge Element Connections

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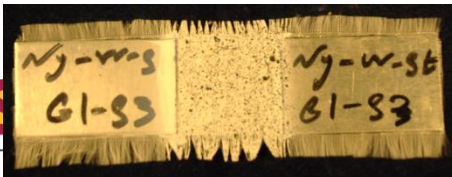
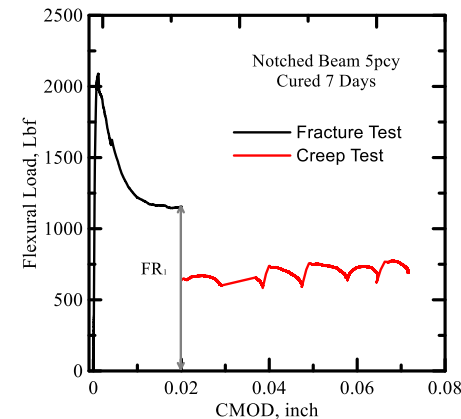
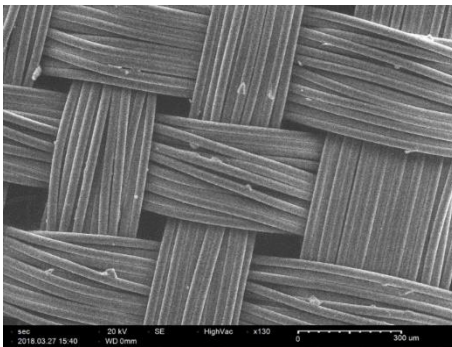
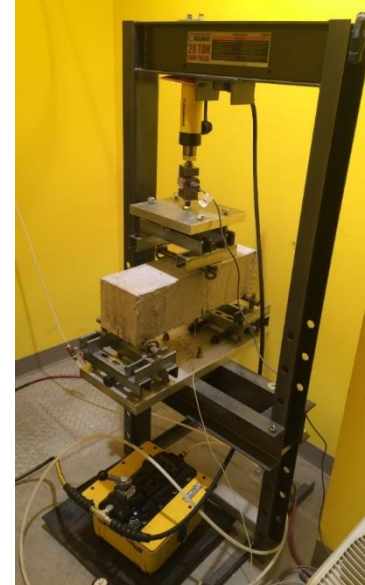
What do we do?

We characterize Materials and Structures

From a single 40 micron fiber all the way to a 4 meter, 9000 lbs tunnel segment (OM=5-8, size)

From High speed tests that take 10 ms, to creep tests that take several months (OM=11, time)

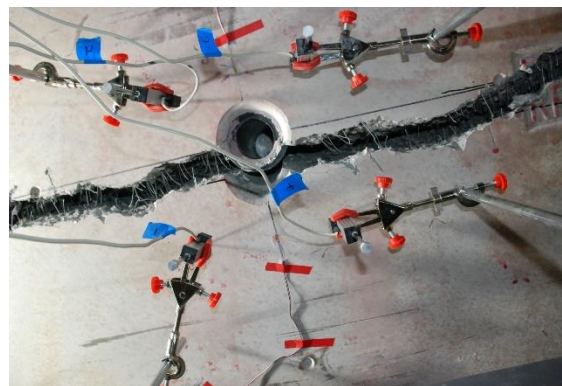
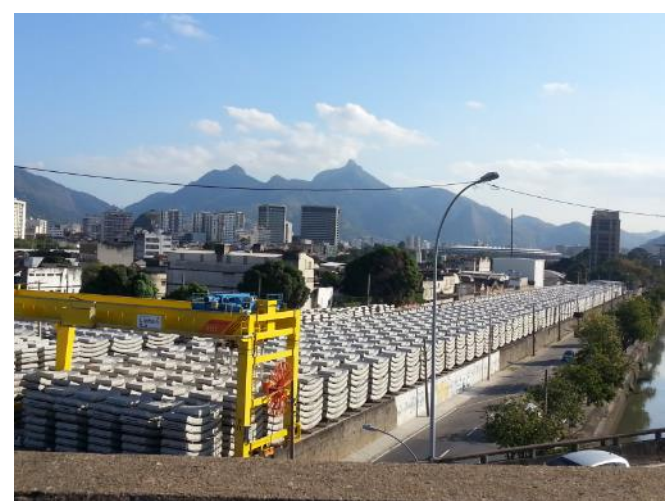
From NASA parachutes for Mars landing to concrete foundation durability due to sulfate attack (OM= geological to planetary)



OM= order of magnitude

ASU Large Scale Structural Testing Lab

- Develop Sustainable Materials, analyze Solid Mechanics, material formulations, Structural components, and Systems.
 - Full scale testing and Modeling in order to promote innovative and sustainable construction systems.
 - solutions for composite materials for transportation, water treatment facilities, pipes, tunnel lining, thin sections, Structural Shapes
- 1) Up to 200 kips in Bending and 800 kips axial capacity
 - 2) Failure mode modeling, Effect of hybrid reinforcement
 - 3) Design tools for Tension, compression, and flexure.
 - 4) A wide range from a 40 μm fiber to a 4 meter tunnel segment
 - 5) Long term serviceability by addressing permeability, creep, and corrosion.

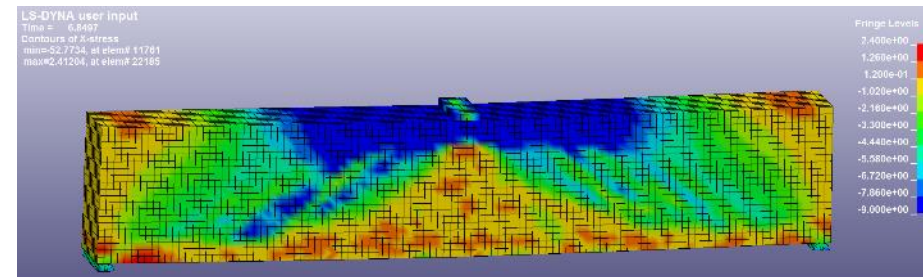
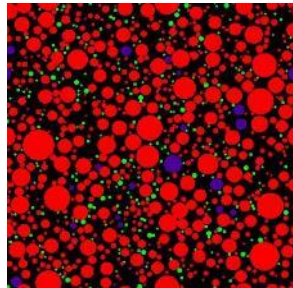
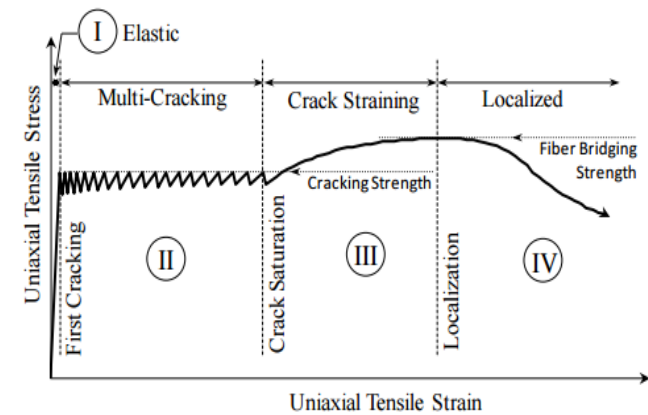


UHPC - Background

- Very high strength and ductility concrete
 - advances in particle packing
 - increased quality control
 - large amounts of fibers
 - very low water-to-cementing materials ratio
- Discontinuous pore structure that reduces liquid ingress, significantly enhancing durability
- Proprietary UHPC mixtures commonly used
- Tends to be very expensive and does not account for local raw materials

Ultra high Performance Concrete Materials

- Designing non-proprietary formulations for UHPC- ADOT
- Utilize Local materials, concrete with 24000 psi strength and 1000 times more ductility
- Particle packing methods and Rheological properties
- Experiments, Design guides, and simulations



Hypothesis: There will be many applications for cost effective non-proprietary UHPC mixtures



Background Information

- Address the potential development of non-proprietary mix designs for manufacturing ultra-high-performance concrete (UHPC) mixtures using materials commercially available in Arizona.
- Several mix designs using all Arizona materials were developed and tested during the study. When tested, they exhibited characteristics that provided more than acceptable performance. Analysis showed these designs to be at a lower cost than UHPC incorporating proprietary materials.
- The cost effectiveness of non-proprietary mixtures would enable ADOT to accelerate bridge construction
- Such mixtures have to be developed and calibrated locally, and none of them have been attempted and tested in Arizona
- ADOT would benefit from economically feasible UHPC mixes specifically designed for Arizona conditions and materials
- ADOT viewed this project as a “**Proof of Concept**” to determine if non-proprietary mixes could perform well and be cost effective

Research Objectives

- Develop UHPC mixtures with local materials that withstand Arizona's varied climatic conditions and meet industry standards for UHPC performance.
- Provide mix design recommendations to develop design criteria for the tested mixtures. The design criteria would be a component of ADOT specifications
- Develop a cost-benefit analysis for the recommended UHPC mix designs versus existing proprietary mixes

Anticipated Benefits for ADOT Groups

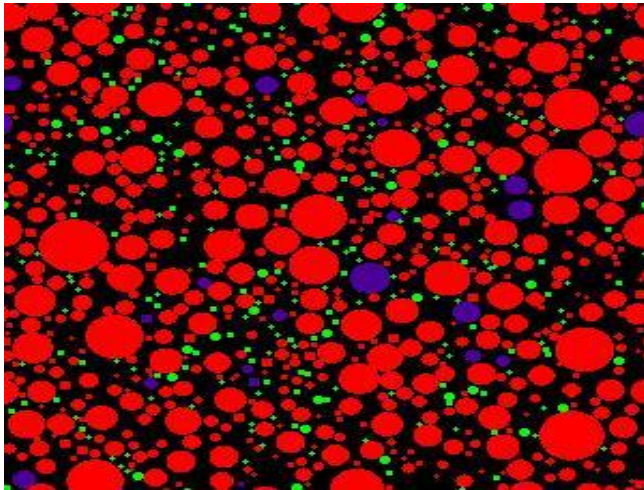
- UHPC mix will accelerate construction time and increase the concrete's durability, strength, ductility, and longevity, making structures available for traffic use faster and producing cost savings
- **Groups involved at ADOT:**
 - ADOT Construction-Materials Group
 - Bridge Group
 - Contracts and Specifications

Main Project Outline

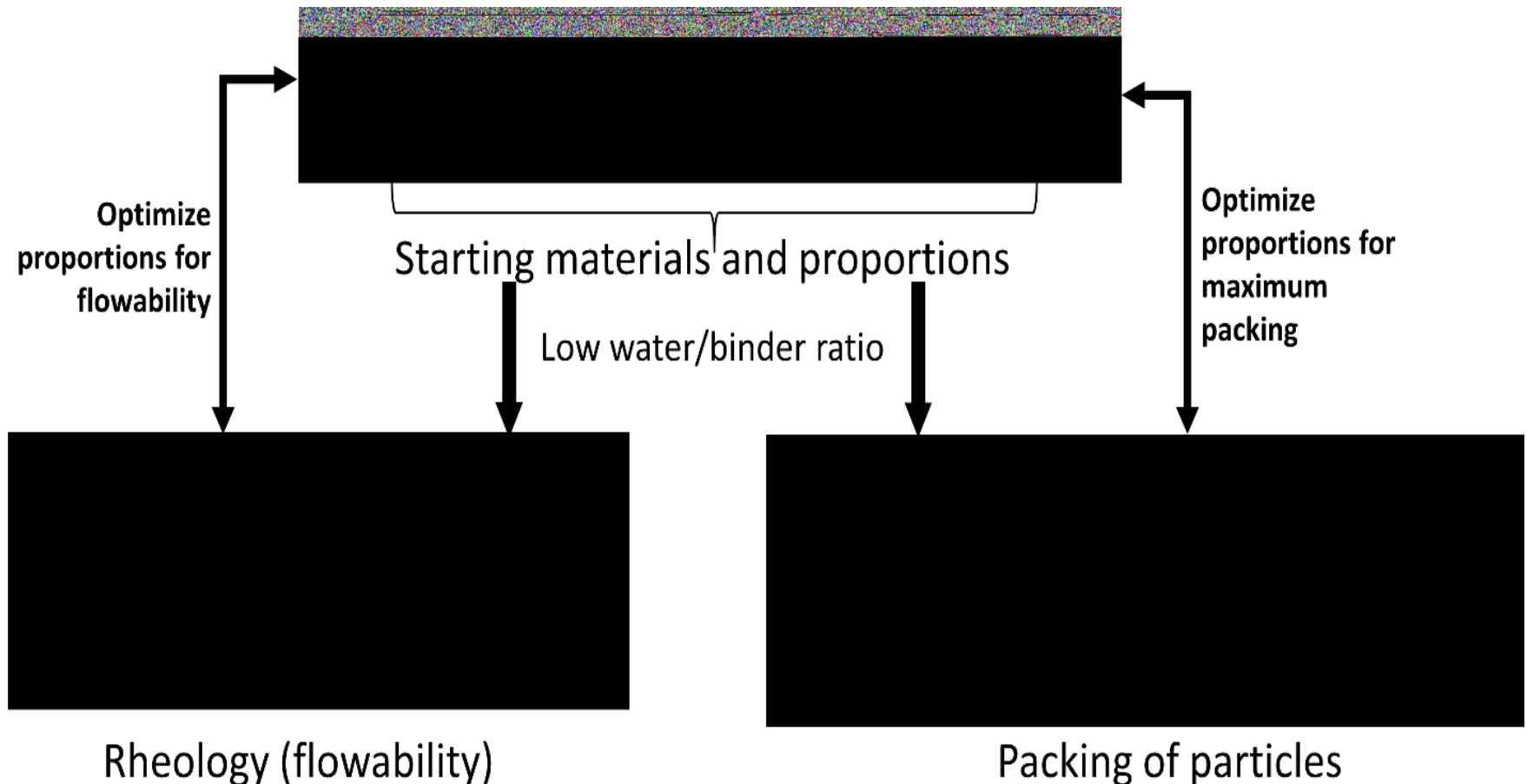
- Introduction
- Microstructure And Rheology Guided Design of Ultra-High Performance Binders Research Objectives
 - Microstructure packing – Selection Criteria and Results
 - Rheology of Pastes – Selection Criteria and Results
 - Characterization of Selected Binders
- Particle Packing Based Design of Ultra-High Performance Concrete
 - Compressible Packing Model
 - Concrete Design Considerations
- Mechanical testing Tension, Compression and Flexure
 - Test methodologies
 - Test results
- Guidelines for Structural Design with UHPC
 - Proposed Methodologies

Task 1: Non-Proprietary Mix Designs

- Microstructure and Rheology Guided Design of Ultra-High Performance Binders
- Designing the ideal paste phase for UHPC to address
 - Local materials and their combinations, low cost
 - Particle packing methods, workability
 - Experiments and simulations
- Rheological properties
- Conducted under a three tier approach, paste, mortar, and concrete

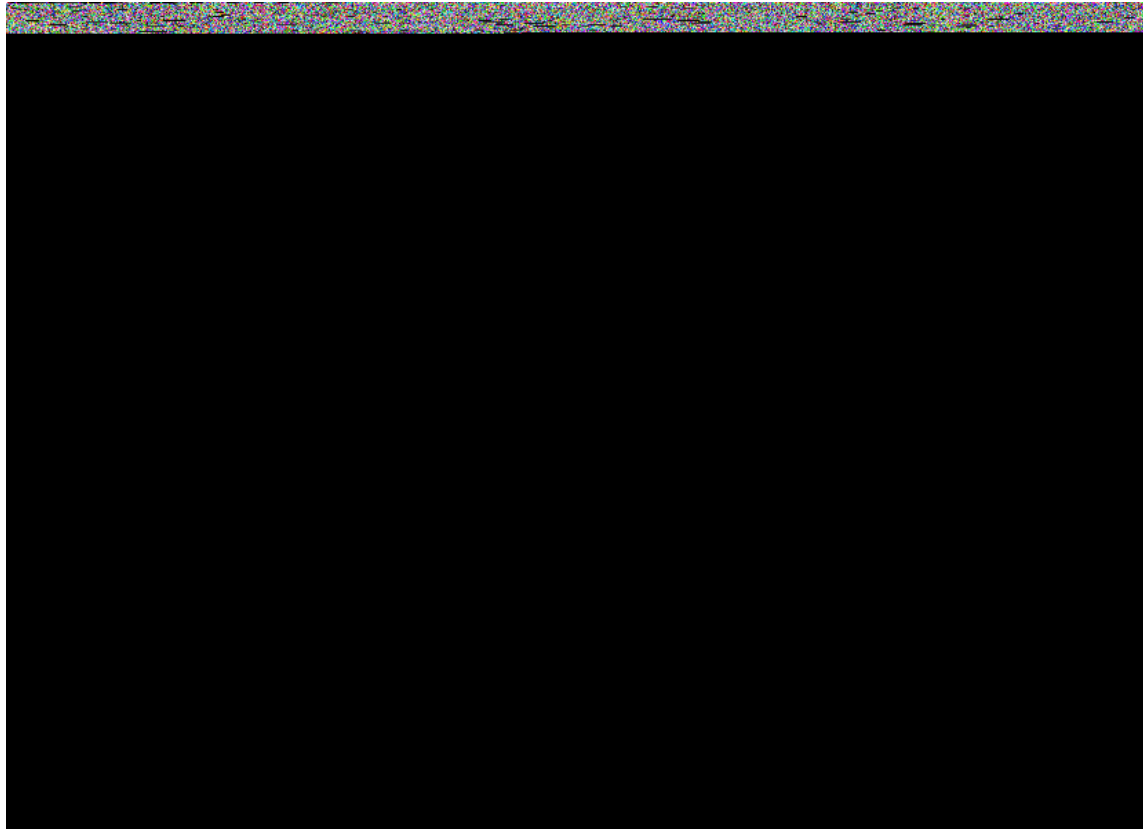


Binder Design based on two metrics of Rheology and Particle Packing



Materials for Paste Selection

- **OPC** – ASTM C150 cement
- **Slag, Silica Fume, Metakaolin** (pozzolanic Silica and Alumina sources – as well as react with carbonates present in the system)
- **Limestone** – 3.0 micron and 1.5 micron. Fine limestone help with dense packing of microstructure.
- **Fly Ash** – pozzolanic, spherical particles aid with workability.



Tertiary and Quaternary Design for Paste

- Blends consisting of Cement, Flyash, Silica Fume, and Slag, Metakaoline, Limestone,
- 33 mixes selected for the study
 - Cement replacement level up to 30% by mass, individual or hybrid combination
 - Water-to-binder ratio of 0.24
 - High-range water reducer content of 1.25% solids content by mass of binder.

Flyash, F
 MicroSilica, M
 Slag, S
 Metakaoline, K
 Limestone, L_a, L_b (3.0
 and 1.5 μm)

K ₁₀	M ₁₀	M ₂₀	F ₂₀	F ₃₀	S ₂₀	S ₃₀	
F ₁₀ K ₁₀	F ₂₀ K ₁₀	S ₁₀ K ₁₀	S ₂₀ K ₁₀	F ₁₀ M ₁₀	F ₂₀ M ₁₀	S ₁₀ M ₁₀	S ₂₀ M ₁₀
F ₂₀ L _{10a}	F ₂₀ L _{10b}	F ₂₅ L _{5a}	F ₂₅ L _{5c}	S ₂₀ L _{10a}	S ₂₀ L _{10b}	S ₂₅ L _{5a}	S ₂₅ L _{5b}
F _{17.5} K _{7.5} L _{5b}	F _{17.5} K _{7.5} L _{5c}	F _{17.5} M _{7.5} L _{5b}	F _{17.5} M _{7.5} L _{5c}	S _{17.5} K _{7.5} L _{5b}	S _{17.5} K _{7.5} L _{5c}	S _{17.5} M _{7.5} L _{5b}	S _{17.5} M _{7.5} L _{5c}

F₂₀ L_{10b} = 20% Flyash and 10% Limestone

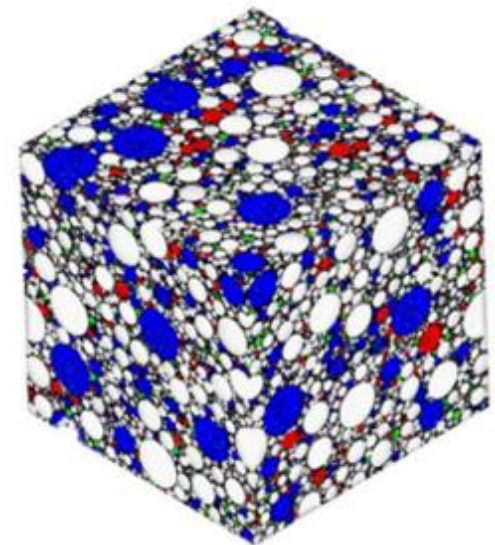
Mix Designs

Mixture composition	Replacement material (% by mass of cement)			
	Fly Ash (F)/ Slag (S)	Metakaolin (K)	Microsilica (M)	Limestone (L); d ₅₀ of 1.5 or 3 μm [#]
UHP-control	0	0	0	0
HP-control	0	0	0	0
OPC + F/S	20, 30	0	0	0
OPC + M	0	0	10, 20	0
OPC + K	0	10	0	0
OPC + F/S + M	10, 20	0	10	0
OPC + F/S + K	10, 20	10	0	0
OPC + F/S + L	20	0	0	10a, 10b
OPC + F/S + L	25	0	0	5a, 5b
OPC + F/S + M + L	17.5	0	7.5	5b,5c
OPC + F/S + K + L	17.5	7.5	0	5b,5c

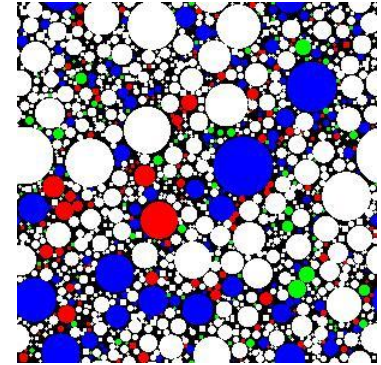
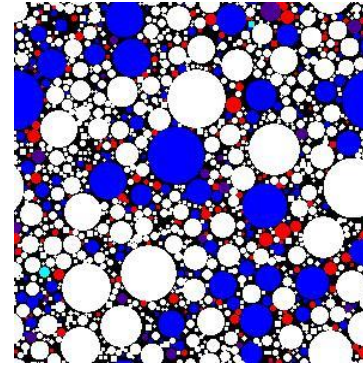
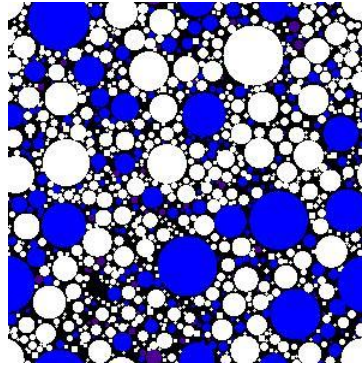
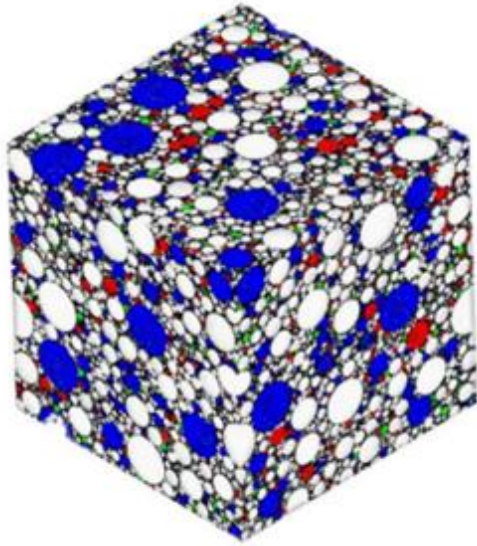
- 33 mixes selected
- Cement replacement, up to 30% by mass
- Water/binder = 0.24
- High-range water reducer (HRWR), .25% solids content by mass of binder

Microstructure Analysis

- ❑ Materials (cement, fly ash, silica fume etc.) have particle sizes ranging from $0.1\text{ }\mu\text{m}$ - $50\text{ }\mu\text{m}$.
- ❑ The smallest unit representing the blend is the representative volume element (RVE).
- ❑ The dimensions of a RVE $300 \times 300 \times 300\text{ }\mu\text{m}^3$.
- ❑ All particles are considered as spherical.
- ❑ MATLAB code generated three-dimensional RVEs for each mixture composition.
- ❑ The problem is similar to the Granular Convection problem, or the Brazil Nut problem.
- ❑ Different properties of the microstructure can be analyzed using the 3D microstructures.



Microstructure Packing



- ❑ 3D volumes are generated using a stochastic particle packing model assuming spherical particles. OPC is in white, fly ash in blue, metakaolin in red and limestone in green.
- ❑ PSDs are discretized to get the number of spheres for each phase.
- ❑ These digitized microstructures are used to obtain key parameters that influence the hydration process.

Microstructural Packing Models– Parameters Selection

Mean Centroidal Distance (MCD)

- Average distance from the center of a particle to the centroid of the microstructure.
- Measure of packing density of the microstructure.

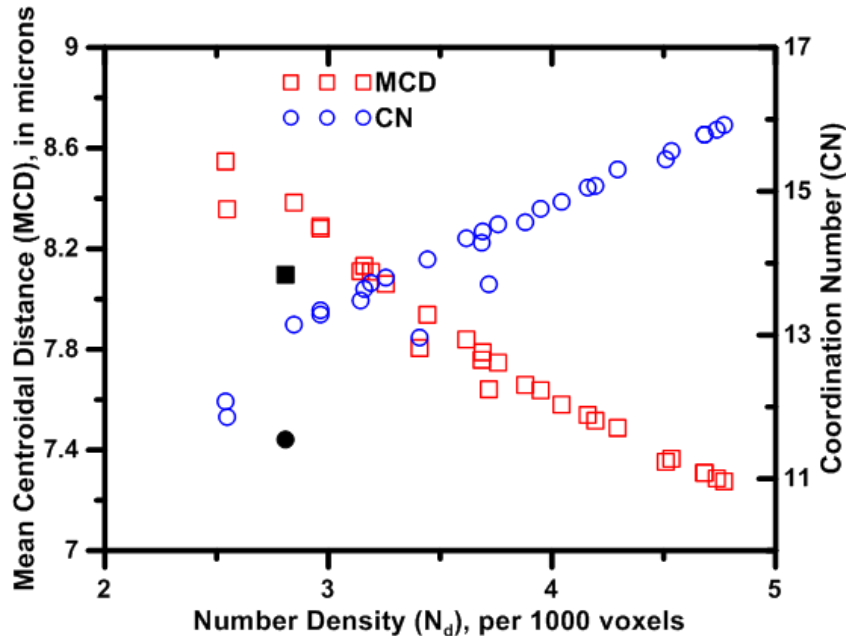
Coordination Number (CN)

- average number of nearest neighbor pairs in the microstructure.
 - For any particle, the nearest neighbor is a particle that lies partially lies in the radial field of that particle, defined as a field with a radius of $(r+\delta r)$ μm , where 'r' is the radius of the particle and δr depends on the average particle size of the mixture.
- represents the inter-particle contacts. It represents packing efficiency.

Number Density (N_d)

- number of particles in a unit volume.
- directly related to the overall powder fineness in the paste. Finer particles, the higher N_d

Microstructural Packing :Results and Selection Criteria



The actual PSD values used for various ingredient components.

As the number density increases, the average particle centroidal distance decreases while the number of particles within the radial distance increases.

You want a high Coordination Number, high Number Density, and low Mean Centroidal Distance

Model 1 :

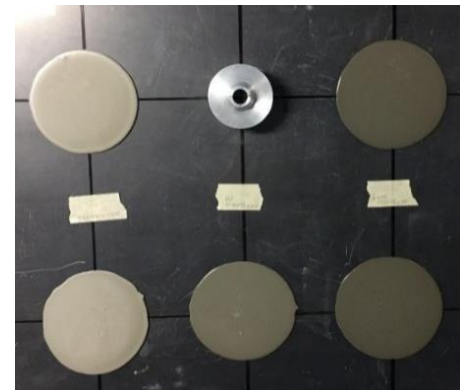
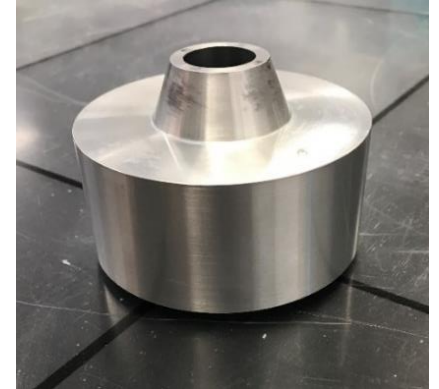
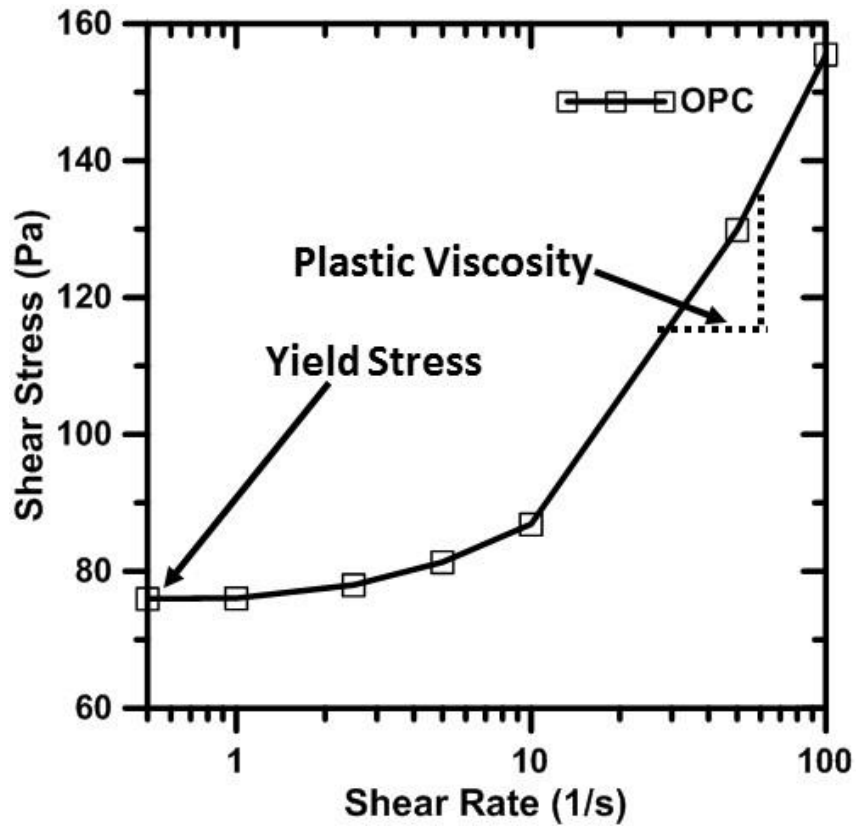
$$\begin{cases} MCD_i \leq MCD_{UHP-control} \\ CN_i \geq CN_{UHP-control} \\ N_{d-i} \geq N_{d-UHP-control} \end{cases}$$

Model 2 :

$$Packing\ Coefficient\ (\gamma) = \frac{CN \times N_d}{MCD}$$

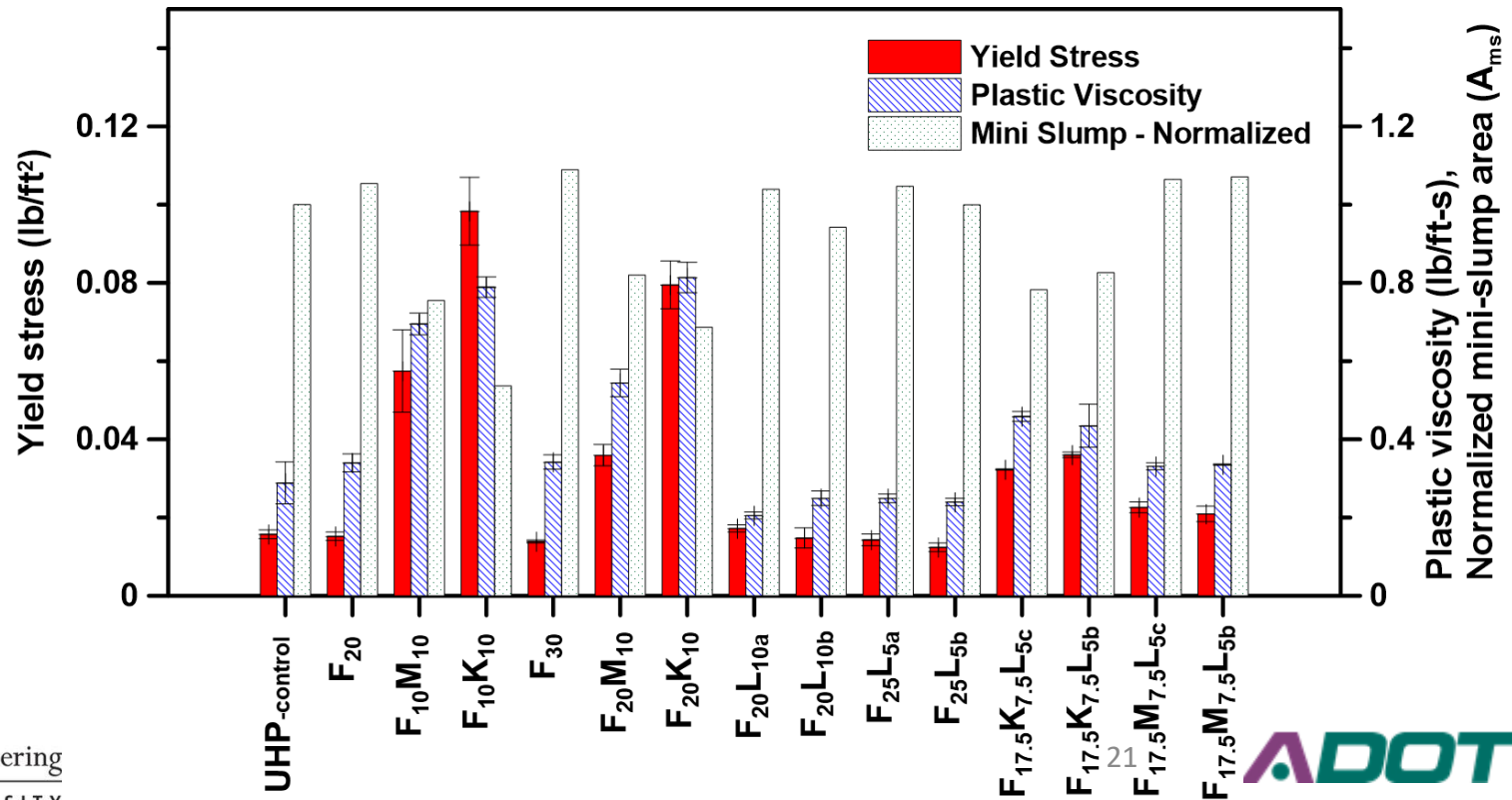
$$\gamma_i \geq 1.0 * \gamma_{UHP-control}$$

Rheology of Pastes – Rheometer Studies and Mini Slump Test



Rheology of Pastes – Rheometer Studies and Mini Slump Test

- the yield stress and plastic viscosity values directly correlate with one another.
- The mini-slump spread is inversely related to the yield stress and plastic viscosity.
- Binary pastes with fly ash have a relatively lower yield stress and plastic viscosity vs. UHP-control paste.
- Interparticle frictional forces are due to particle size and surface characteristics.
- Ternary mixes with metakaolin show high yield stress.
- Fine limestone improves the overall packing, reduces the inter-particle friction in the quaternary mixtures, lowers the yield stress and plastic viscosity, thereby leading to increased flowability of these mixtures.

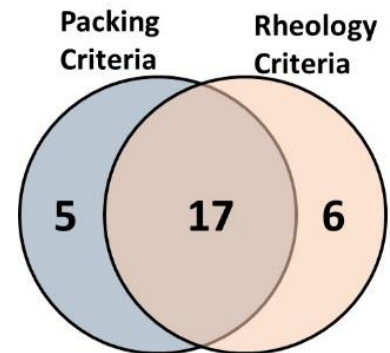
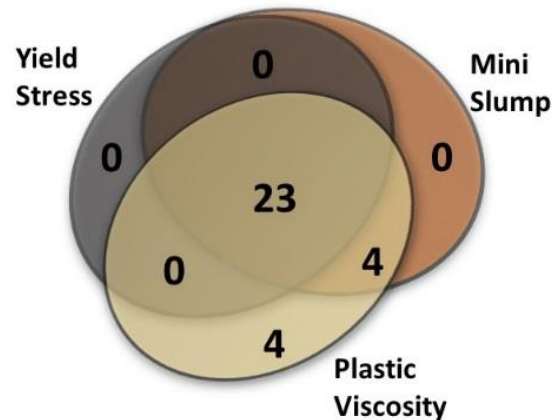
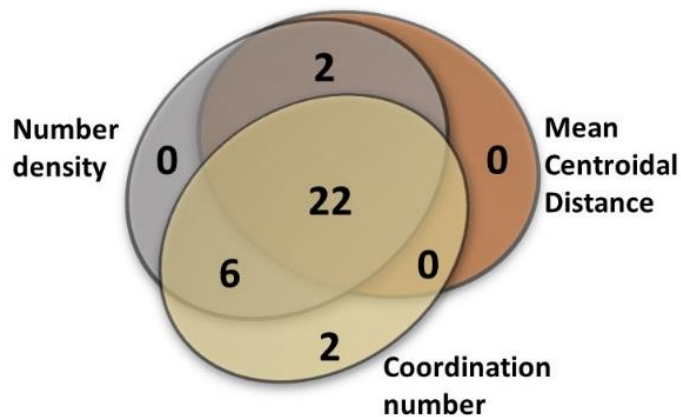


Rheology of Pastes – Selection Criteria

- Require the yield strength and plastic viscosity of the paste be sufficiently low by using high amounts of HRWR.
- Start with a very low rheology parameter and then increase the particle packing to a workable level.
- Even 2-to-3 times higher yield stress and/or plastic viscosity than that of the UHP-control would not detrimentally influence the workability.

Selection based on Overall Analysis

- Start with a very low rheology parameter and then increase the particle packing to a workable level. Even 2-to-3 times higher yield stress and/or plastic viscosity than that of the UHP-control would not detrimentally influence the workability.
- Out of all the 33 mixes planned, 22 match the theoretical packing criteria and 23 matched the experimental rheology criteria
- 17 mixes matched both rheology and packing criteria
- 8 mixtures were further chosen among these candidates for strength studies



Mixtures Selected

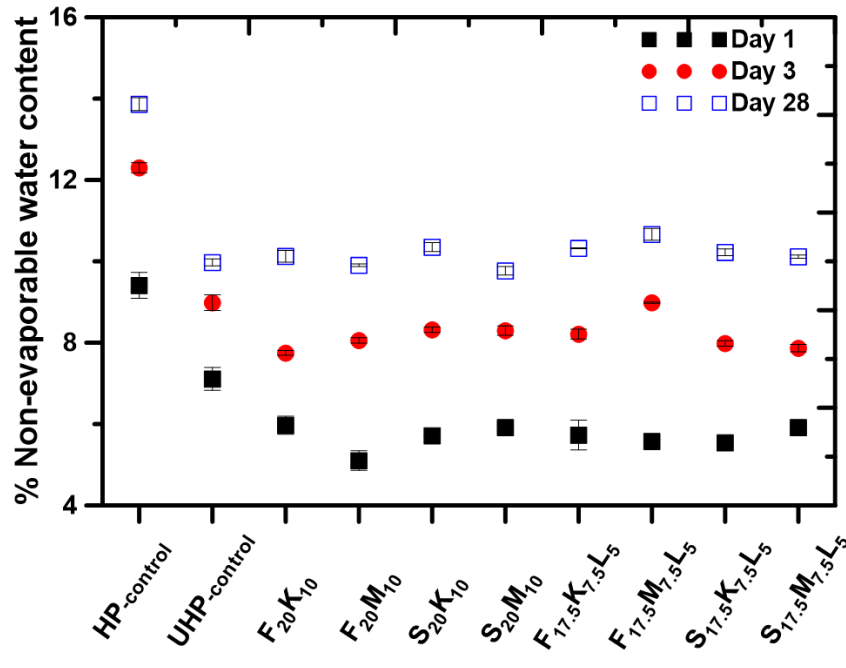
K_{10}	M_{10}	M_{20}	F_{20}	F_{30}	S_{20}	S_{30}	
$F_{10}K_{10}$	$F_{20}K_{10}$	$S_{10}K_{10}$	$S_{20}K_{10}$	$F_{10}M_{10}$	$F_{20}M_{10}$	$S_{10}M_{10}$	$S_{20}M_{10}$
$F_{20}L_{10a}$	$F_{20}L_{10b}$	$F_{25}L_{5a}$	$F_{25}L_{5c}$	$S_{20}L_{10a}$	$S_{20}L_{10b}$	$S_{25}L_{5a}$	$S_{25}L_{5b}$
$F_{17.5}K_{7.5}$ L_{5b}	$F_{17.5}K_{7.5}$ L_{5c}	$F_{17.5}M_{7.5}$ L_{5b}	$F_{17.5}M_{7.5}$ L_{5c}	$S_{17.5}K_{7.5}$ L_{5b}	$S_{17.5}K_{7.5}$ L_{5c}	$S_{17.5}M_{7.5}$ L_{5b}	$S_{17.5}M_{7.5}$ L_{5c}

Mixtures in the red shaded cells were chosen for detailed studies (other shaded in green also met the criteria but were not used). Since good packing required microsilica or metakaolin, all the mixes were required to use one of those components

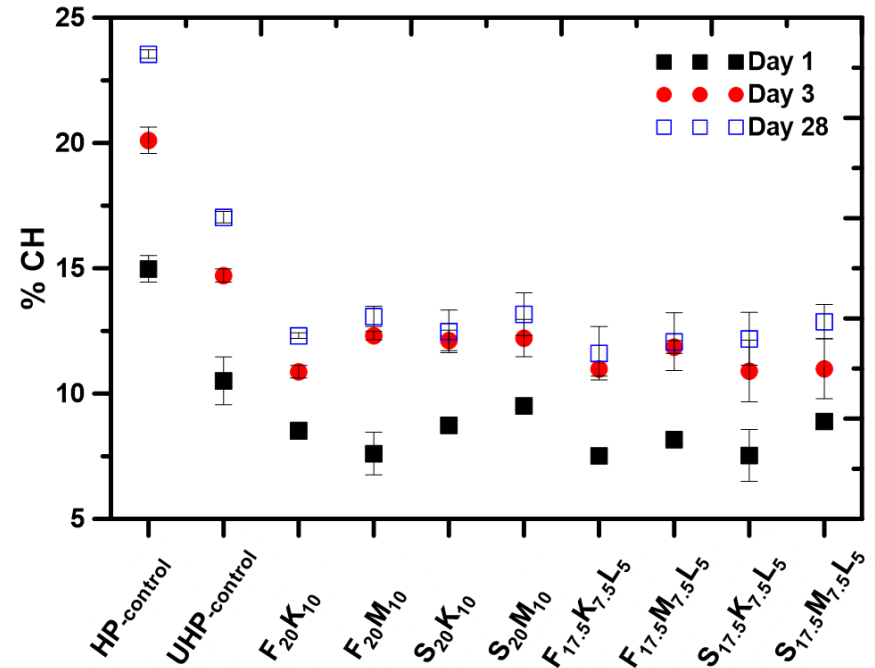
A Arora, et. al. " Microstructural packing and rheology-based binder selection and characterization for Ultra-high Performance Concrete (UHPC)" , Cement and Concrete Research 103, 179-190, 2018



Characterization of Selected Binders based on cumulative particle reactivity (degree of hydration)



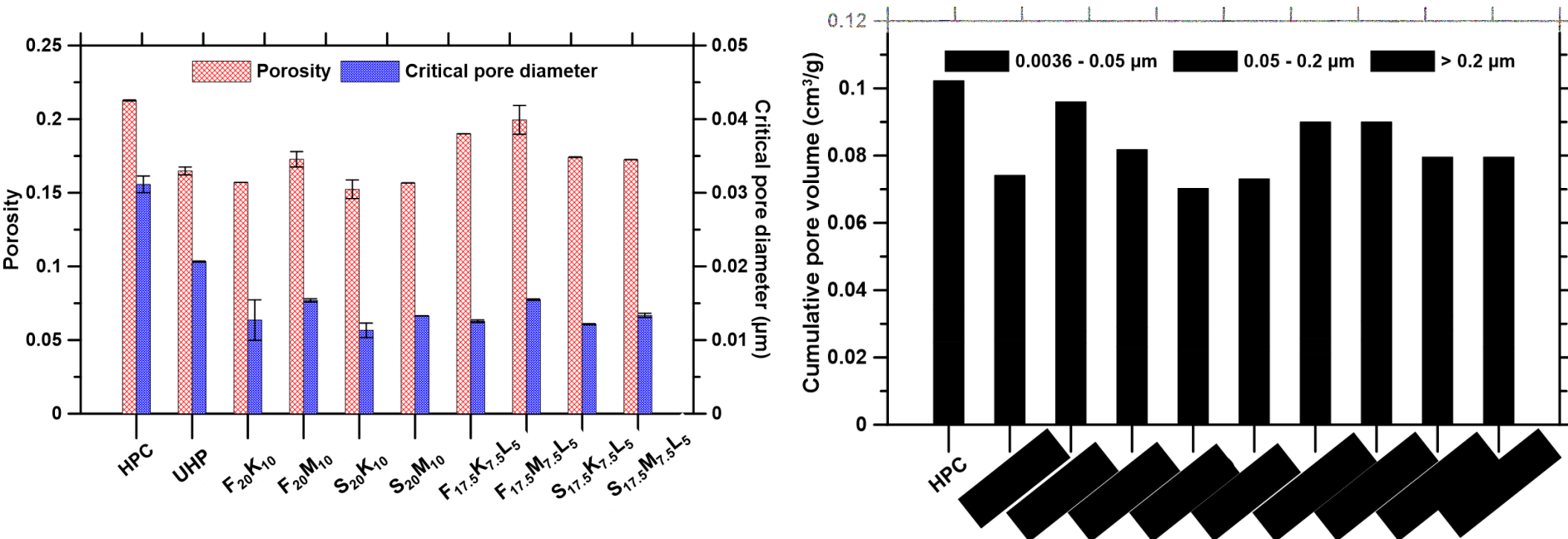
- Non-evaporable water content increases with age of sample.
- indicative of the degree of hydration of the paste.



- CH content remains constant from Day 3 to Day 28.
- indicative of the degree of pozzolanic reaction in the pastes

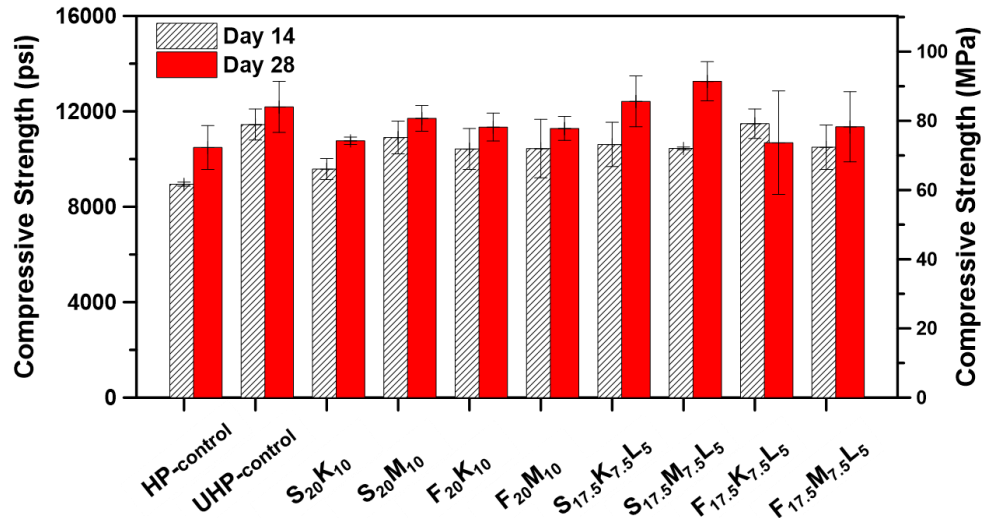
Characterization of Selected Binders –Durability Metrics

Porosity and pore Structure

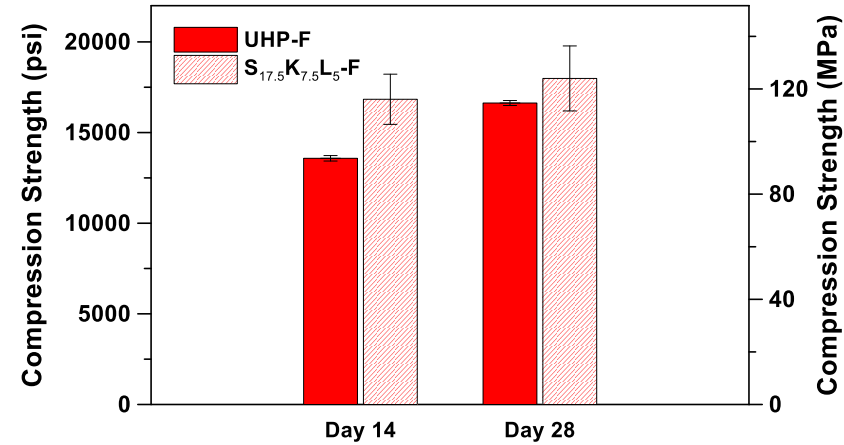


- Ternary blends show overall porosities similar to that of UHP-control mixture.
- Quaternary blends show slightly higher porosities than the ternary blends, however these mixes have a higher proportion of pores present in the 0.0036-to-0.05 μm range.
- High pore volume in the hydration products (i.e., gel pores), indicates increased formation of reaction products
- making smaller pores due to better packing and increased reactivity

Characterization of Selected Binders – Compressive Strength



Strength without steel fibers



Strength with 2.5% steel fiber content

- Most of the UHP mortars demonstrate 28 day strengths comparable to, or higher than that of the control UHP mortar, even with a clinker factor reduction of 30%.
- Incorporation of steel fibers resulted in a 28 Day compressive strength of ~125 MPa.

Paste Mix Designs for Mortar framework

- Mixtures that obtained the maximum compressive strength values were selected.
- Water/cement ratio was reduced from initial value of 0.24 to 0.20 or less.
- This was achieved by optimization of water to superplasticizer ratio to have more water in the mixture while maintaining the same workability.
- Flyash +Metakaolin vs. Slag+ Silica Fume, plus limestone
- Mixing at high shear rate for longer duration.

Mixture composition	Mixture ID	Replacement material (% by mass of cement)			
		Fly Ash (F)/ Slag (S)	Metakaolin (K)	Microsilica (M)	Limestone (L); d_{50} of 1.5 or 3 μm
OPC + M + L	T_SF	0	0	20	30
OPC + S + M + L	Q_SL	17.5	0	7.5	5b,5c
OPC + F + K + L	Q_FA	17.5	7.5	0	5b,5c

Limestone:

'a' – 1.5 μm , 'b' – 3 μm
 'c' – 50% 1.5 μm + 50% 3 μm

Mixing Procedure

A number of mixing procedures were employed including the use of high shear Omni mixer, Hobart mixer and hand-held Dewalt spade-handle drill.

Step 1 - Aggregate + Silica Fume

Step 2 – Add OPC and Fly Ash

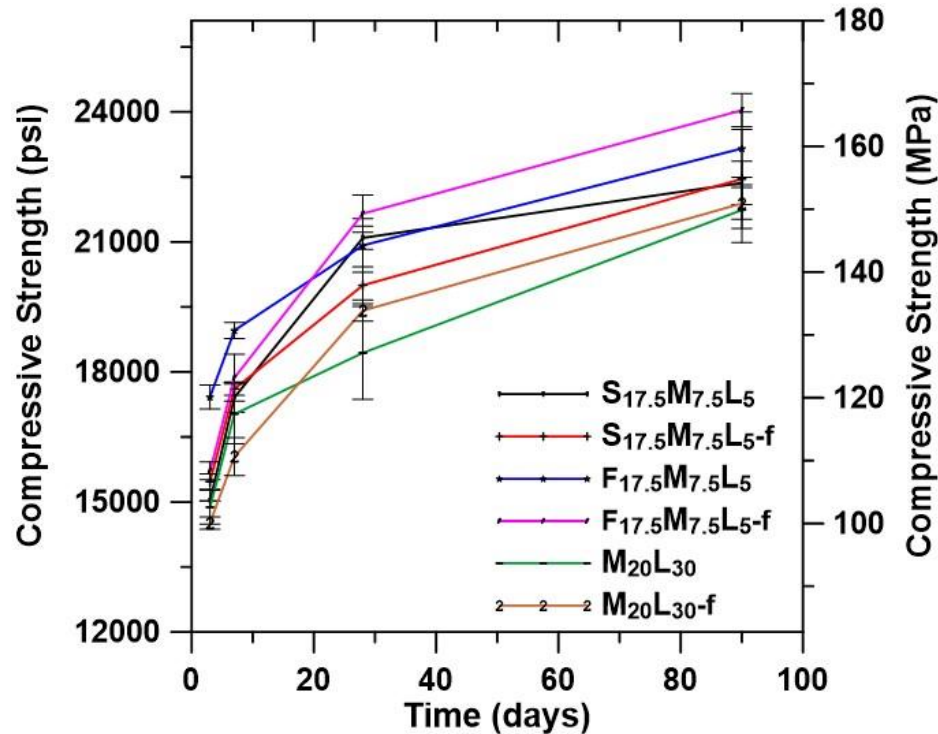
Step 3 – Add water and superplasticizer in increments until a cohesive mixture is obtained.

Step 4 – Add fibers

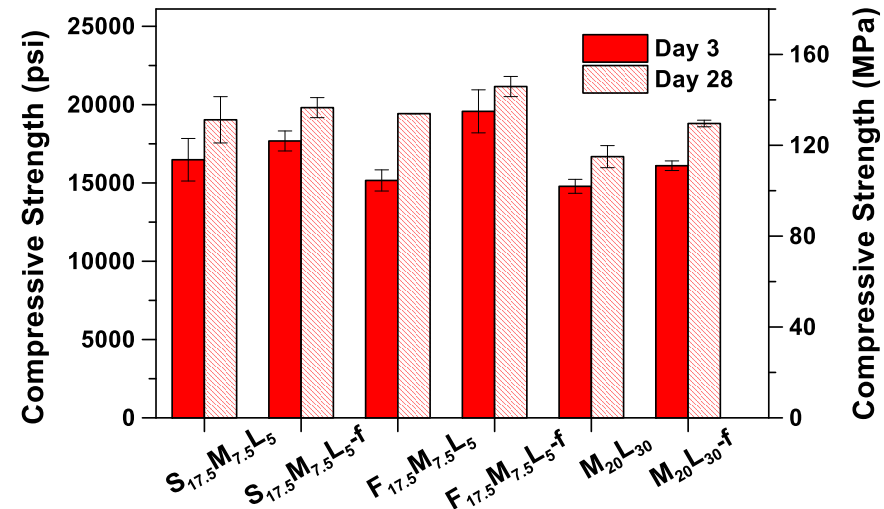


Mortar Strength Results

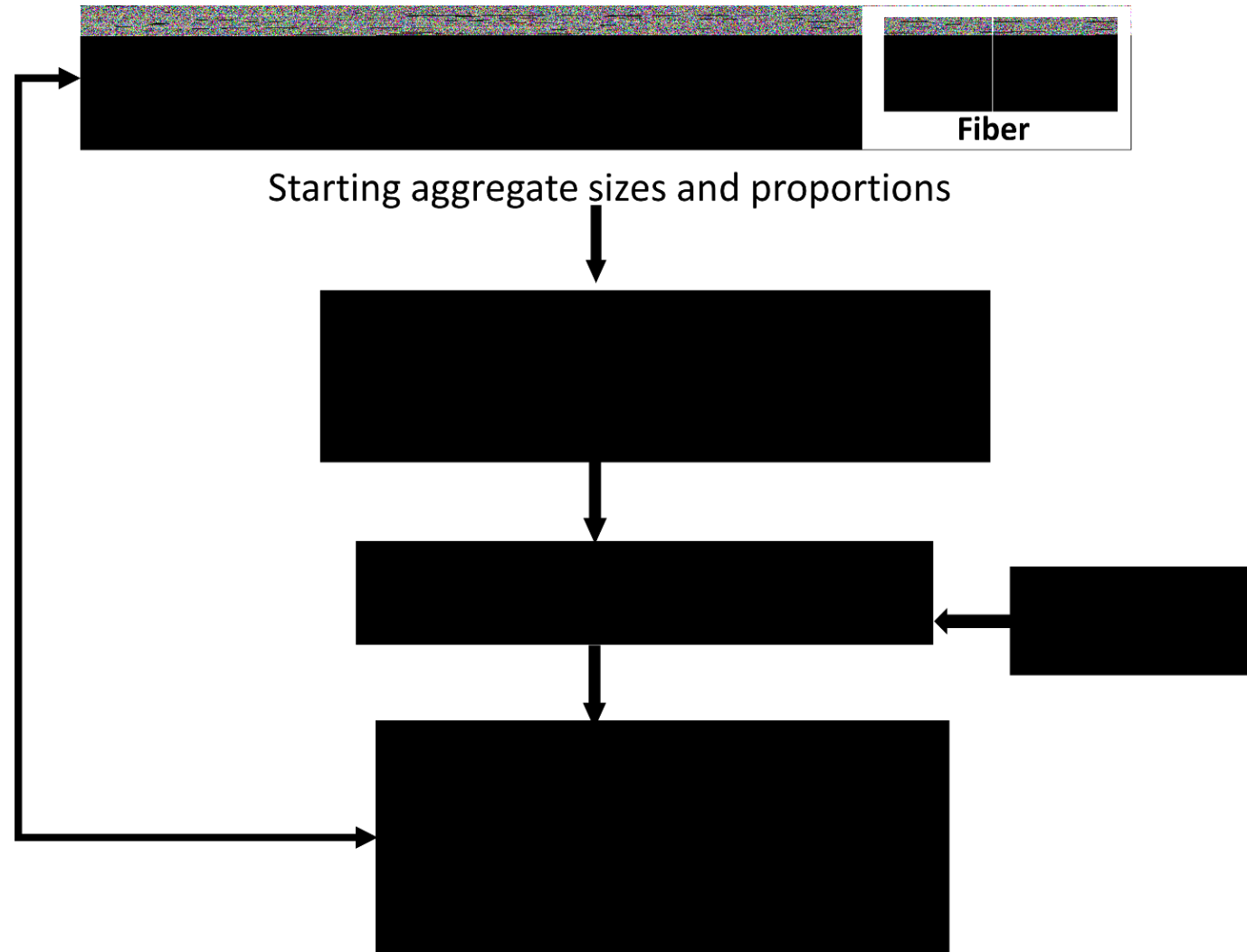
Compressive strength results for
75 mm x 150 mm cylinders



Compressive strength results for 50
mm cubes – Heat cured at 80°C



UHPC Concrete Design



Aggregate Classes Used

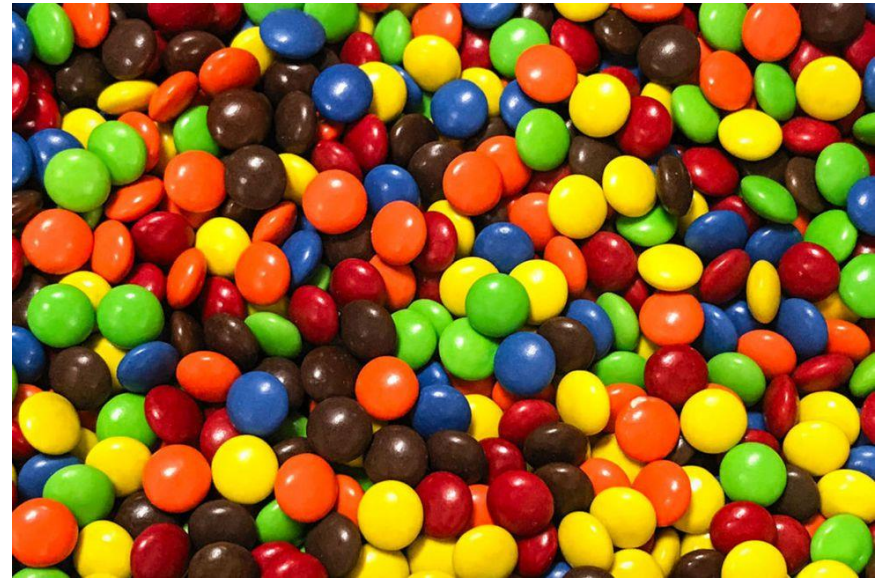
- 5 different aggregate classes were used corresponding to sizes - #4, #8, #10, coarse sand with a $d_{50} = 0.6$ mm, fine sand with a $d_{50} = 0.2$ mm
- Steel fibers – $d = 0.6$ mm, $l = 13$ mm.



Mechanical Splitter used to obtain uniform gradation of particles

Compressible Packing Model

- Calculates the **random virtual packing density** based on the particle size distributions of the constituents.
- Uses a **compaction index 'K'** to account for the compaction process and evaluates the actual packing density, Φ .
- It is assumed that mixtures are composed of several components of equal-density particles. **Particles may not be spherical particles.**



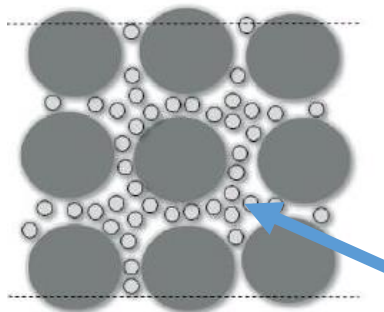
Compressible Packing Model – Wall and Loosening Effect

$$\gamma_i = \frac{\beta_i}{1 - \sum_{j=1}^{i-1} [1 - \beta_i + b_{ij}\beta_i(1 - 1/\beta_j)]y_j - \sum_{j=i+1}^n [1 - a_{ij}\beta_i/\beta_j]y_j}$$

$$a_{ij} = \sqrt{1 - (1 - d_j/d_i)^{1.02}}$$

a_{ij} - Loosening effect coefficient

- Reduction in packing density of large sized grains due to the presence of smaller sized grains

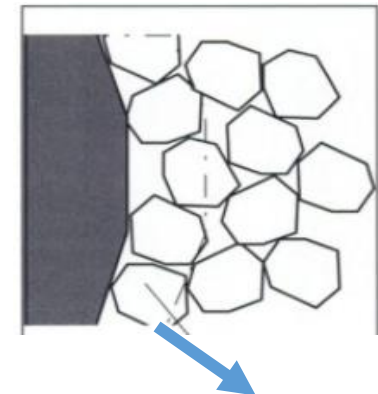


Loosening Effect

$$b_{ij} = 1 - (1 - d_j/d_i)^{1.50}$$

b_{ij} - Wall effect coefficient

- Represents the reduction in packing density of smaller grains due to the presence of a relatively larger grain.



Wall Effect

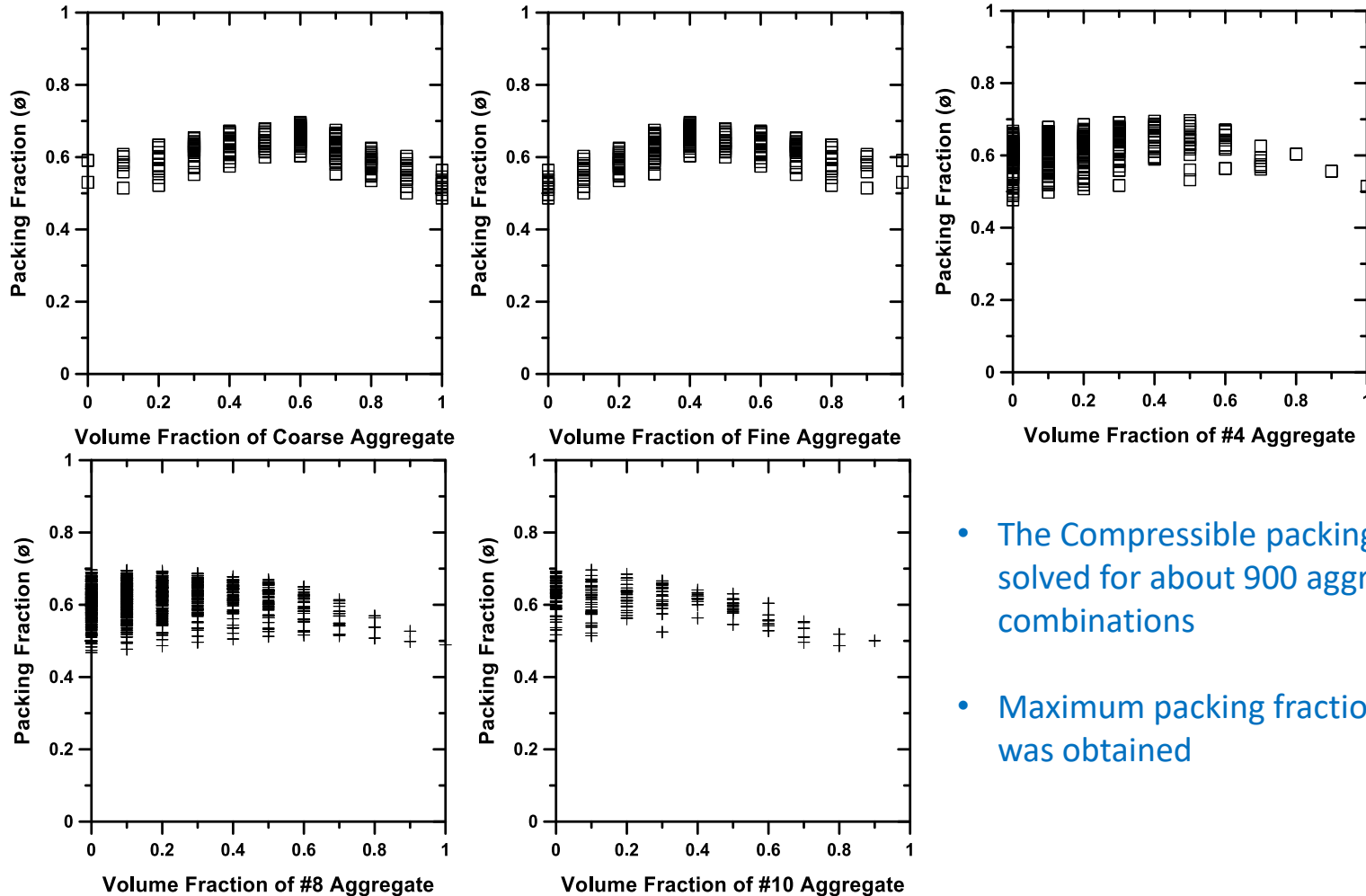
Aggregates – Measured Packing Values

Aggregate Size	#4	#8	#10	Concrete Sand	Fine Sand
Packing Density	0.572	0.544	0.520	0.620	0.527
Average Diameter	0.25"	0.09"	0.08"	0.02"	0.009"



DRUW Method to determine
packing density of aggregates

CPM - Results

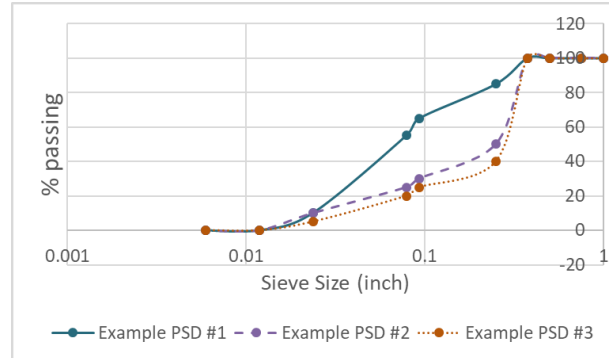


- The Compressible packing model was solved for about 900 aggregate combinations
- Maximum packing fraction of 0.696 was obtained

Compressible Packing Model (CPM)



5 different aggregate classes were used corresponding to sizes - #4, #8, #10, coarse sand with a $d_{50} = 0.6$ mm, fine sand with a $d_{50} = 0.2$ mm



Compressible Packing Model

Random Packing Density

$$\begin{aligned}\Phi(\text{example psd \#1}) &= \Phi_1 \\ \Phi(\text{example psd \#2}) &= \Phi_2 \\ \Phi(\text{example psd \#3}) &= \Phi_3\end{aligned}$$

Results from CPM

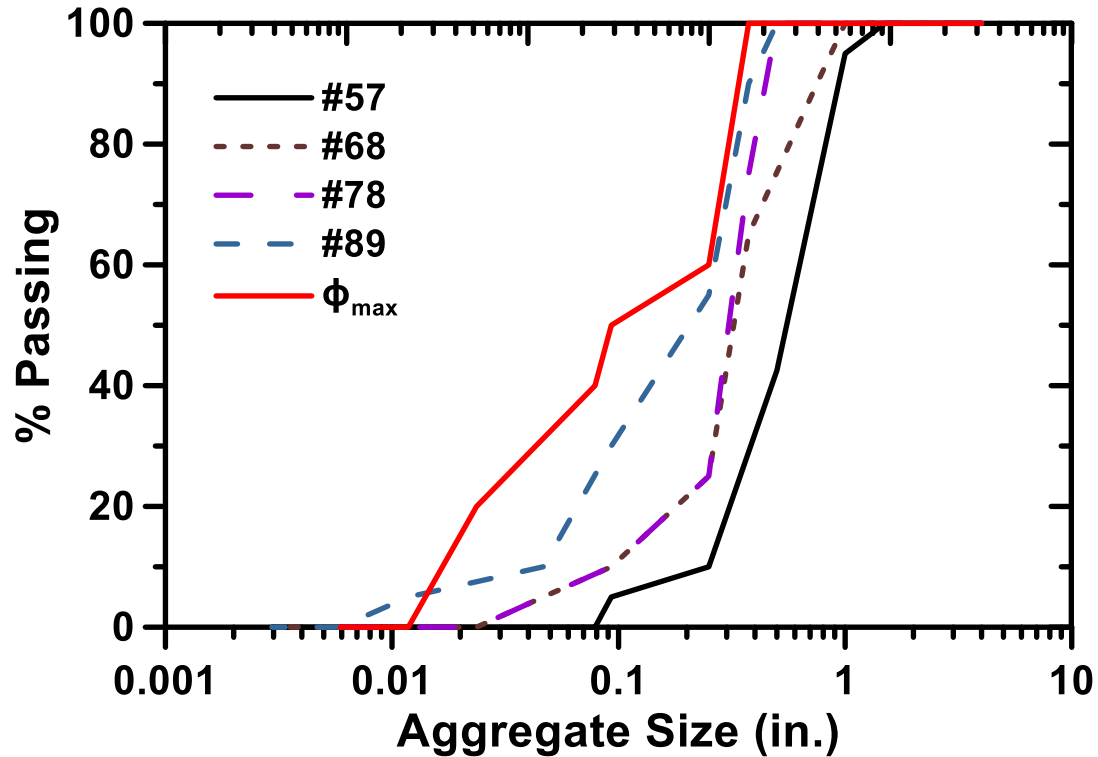
Aggregate
Combination for
maximum packing
density



#4	#8	#10	Concrete Sand	Fine Sand	Packing density
0.4	0.1	0.1	0.2	0.2	0.696
0.3	0.1	0.2	0.2	0.2	0.685
0.1	0.3	0.1	0.3	0.2	0.675
0.5	0.1	0.1	0.2	0.1	0.671
0.1	0.3	0.2	0.3	0.1	0.657
0.2	0.1	0.4	0.1	0.2	0.634
0.1	0.1	0.1	0.2	0.5	0.599
0	0	0	0.6	0.4	0.591
0.2	0.1	0.5	0.1	0.1	0.587
0.4	0.1	0.5	0	0	0.546
0.1	0.1	0.8	0	0	0.494

- Each combination of aggregates results in a unique value for packing density
- Maximum packing fraction of 0.696 was obtained

CPM – Cumulative Size Distribution of Aggregates

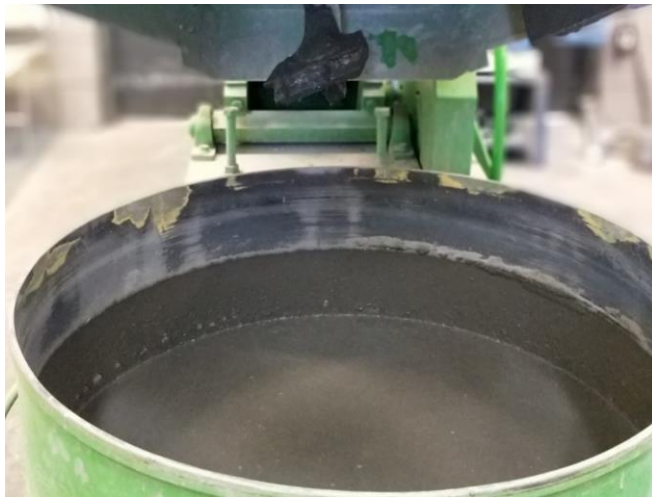


Aggregate Type	#57	#68	#78	#89	ϕ_{max}
Packing Density	0.600	0.612	0.613	0.655	0.696

Implementation Plan

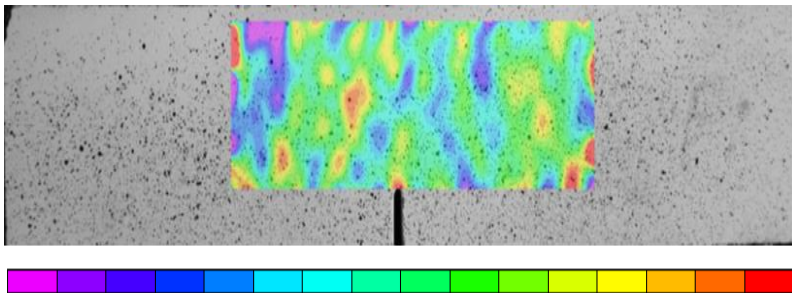
- Water to binder ratio of 0.16 – 0.18 (w/w), Aggregate to binder ratio of 0.7 by mass
- W/b depends on the workability of the mixture.
- Optimized aggregate proportions
- Superplasticizer content of 1.25% solids by weight of binder.
- Steel fibers – 1 and 3% by volume of mixture, when added.

#4	#8	#10	Concrete Sand	Fine Sand	Packing Fraction
0.4	0.1	0.1	0.2	0.2	0.696

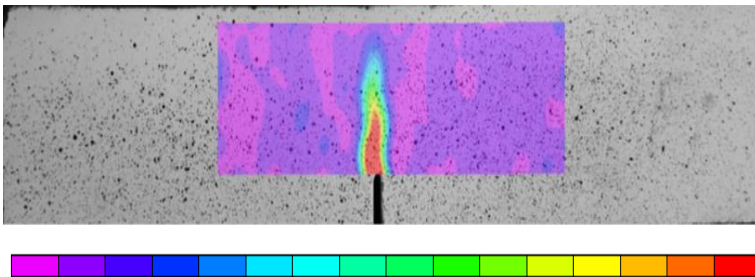


Task 2: Testing of UHPC concretes

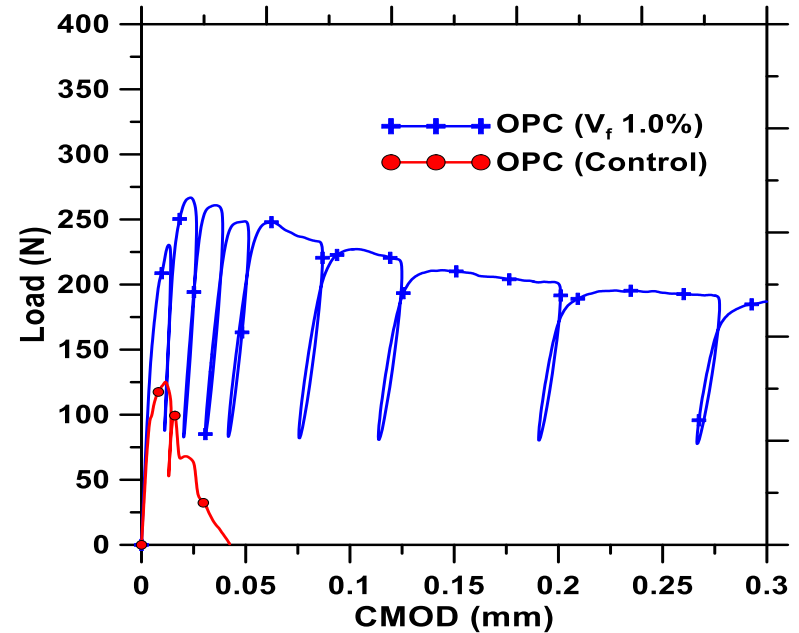
- Mixing and placing of UHPC
- Strength and modulus development
- Toughness/Ductility



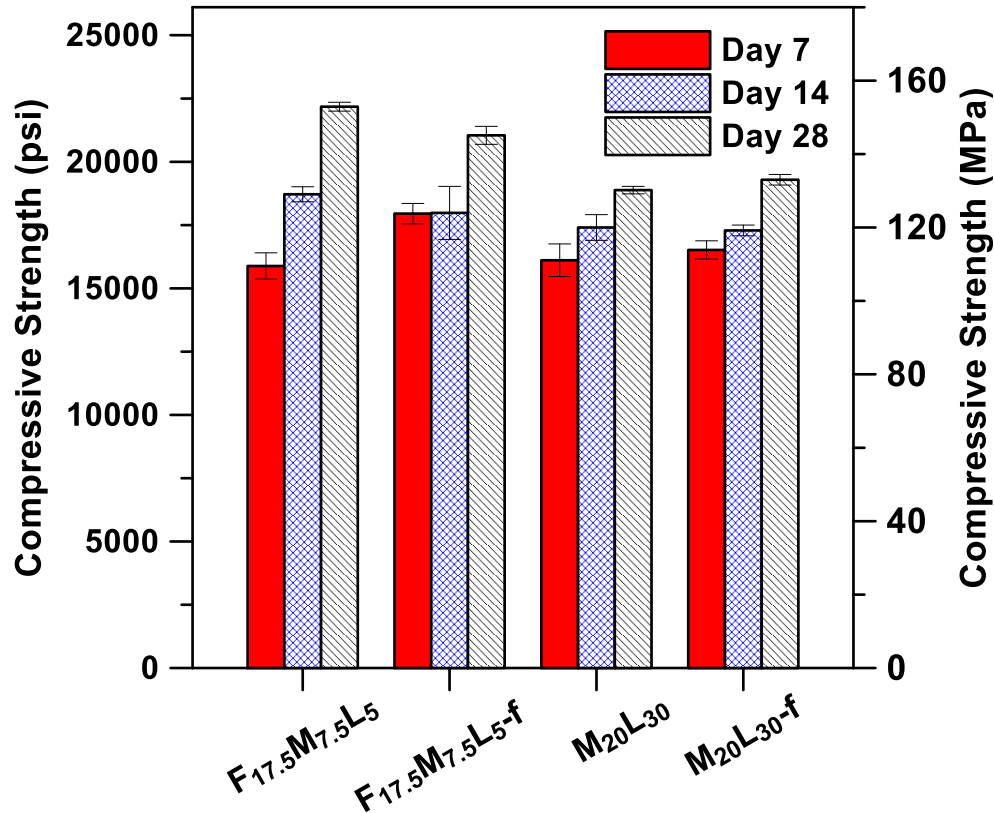
0.0000 0.0002 0.0004 0.0006



0.0003 0.0008 0.0013 0.00175



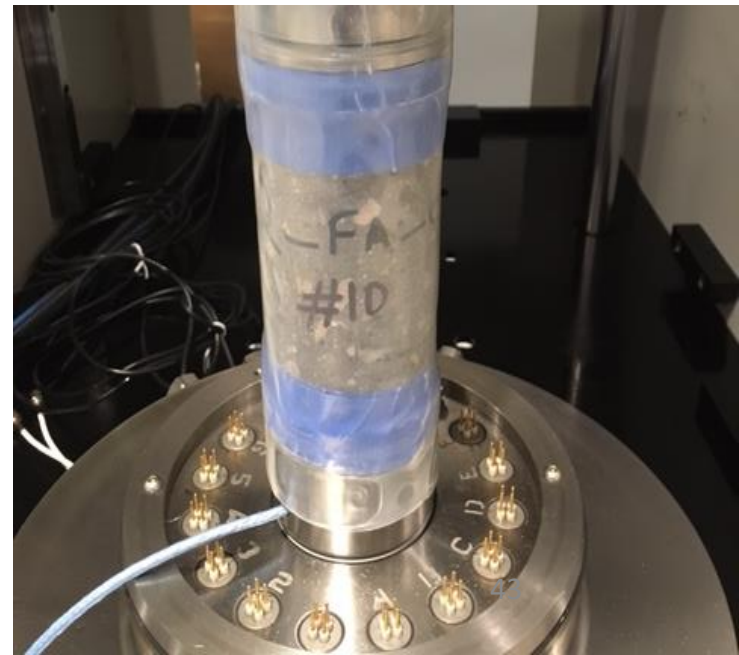
Compression Strength of Concrete Mixtures



- Compressive strength was evaluated at 7, 14 and 28 days of hydration using a 600 kip load frame.
- 28 day strengths of at least 130 MPa were obtained for all concrete mixtures.

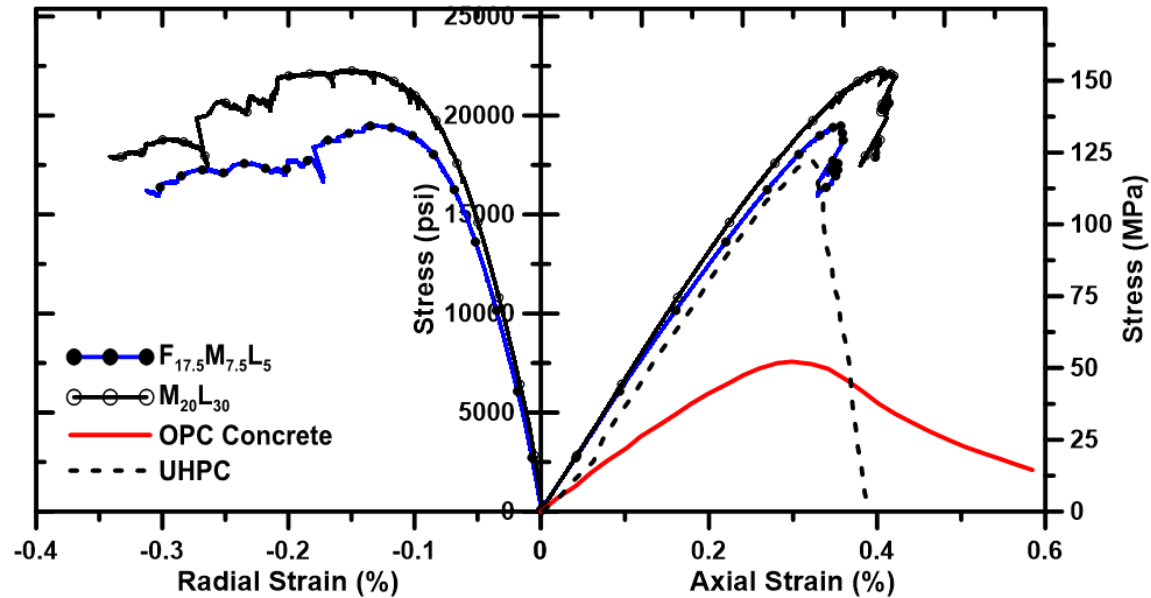
Stress-Strain Response in Compression – Test Setup

- 2" x 4" cylindrical specimens were cored from 3" x 6" concrete specimens to be used for the evaluation of stress-strain response.
- Ends of the cylinders were ground to extremely low surface roughness (< 0.007 inches).
- In-situ ultrasonic pulse velocity (UPV) testing was also conducted during the compression test and velocity measurements were recorded at successive stress intervals of 10 MPa.

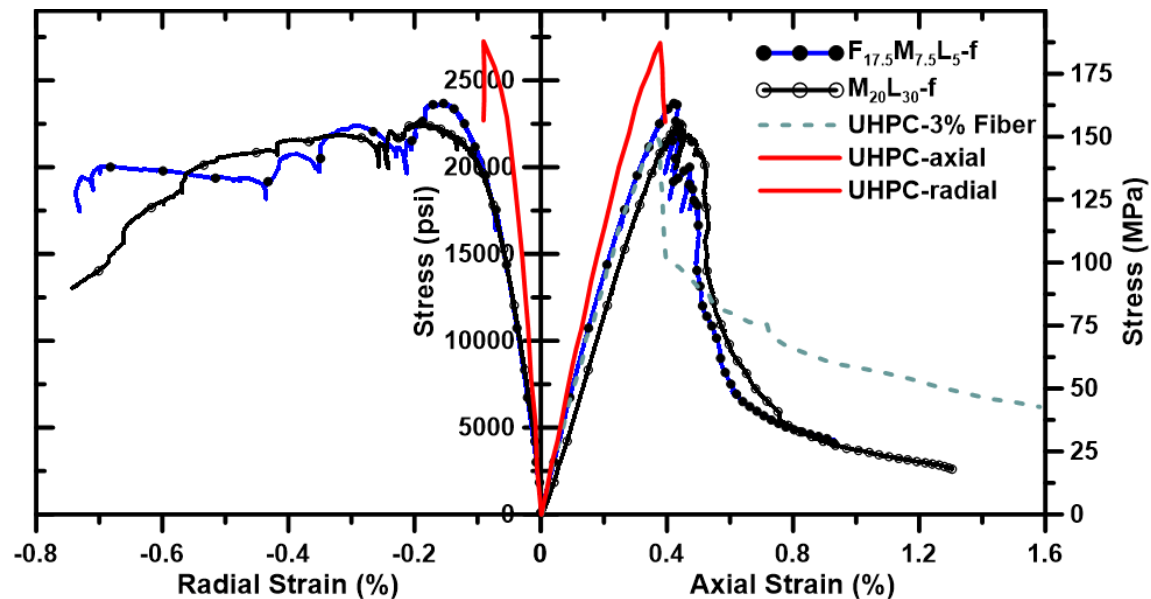


Stress-Strain Response in Compression

Mixtures without fibers

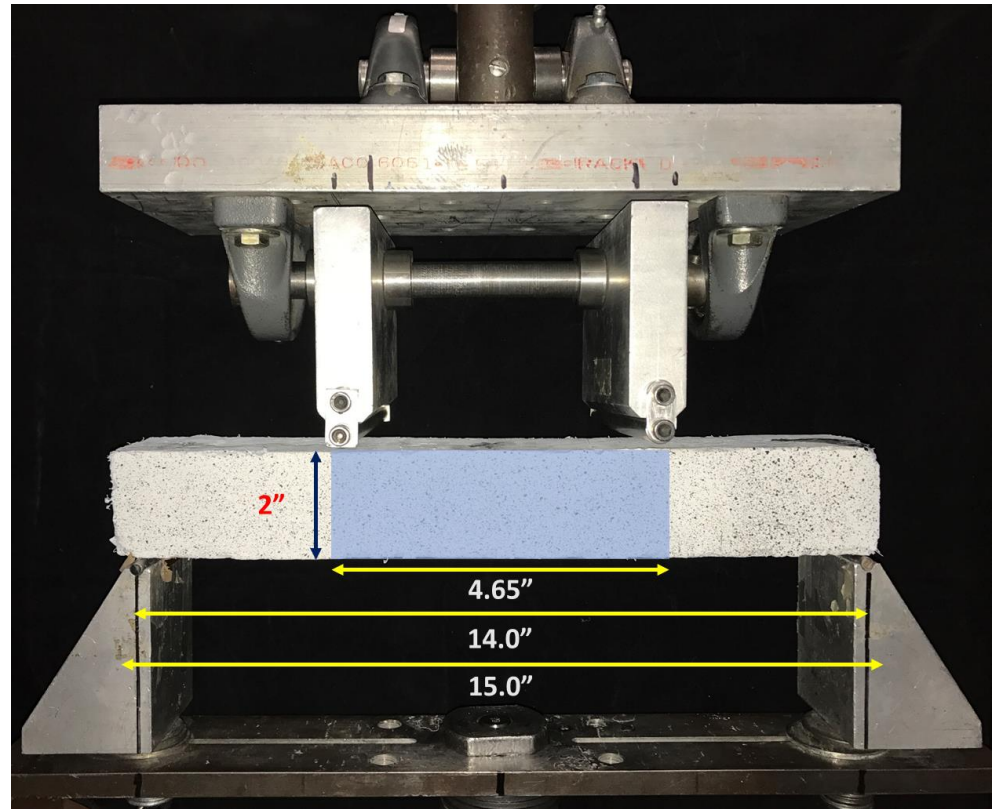


Mixtures with fibers

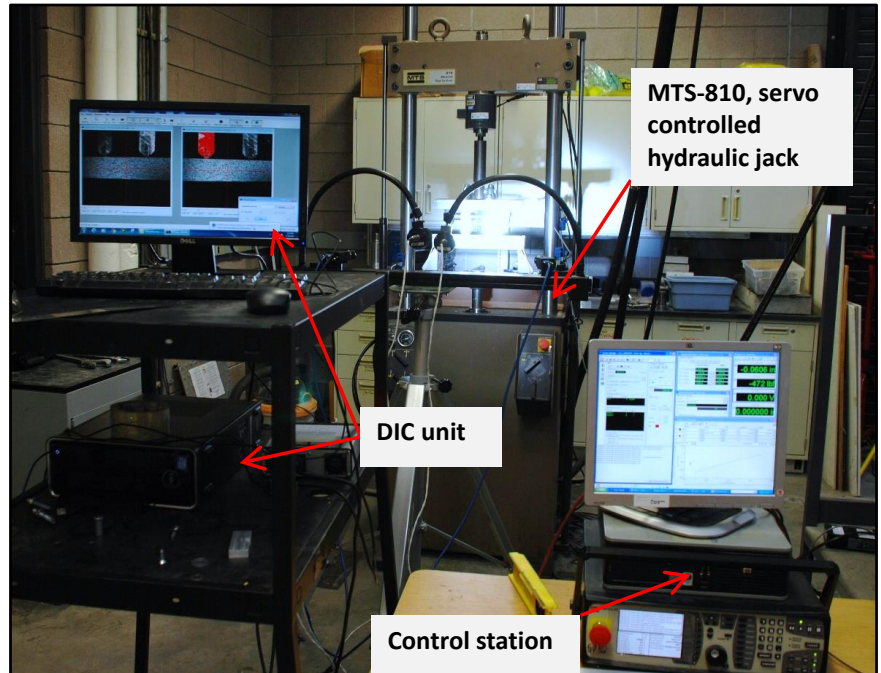
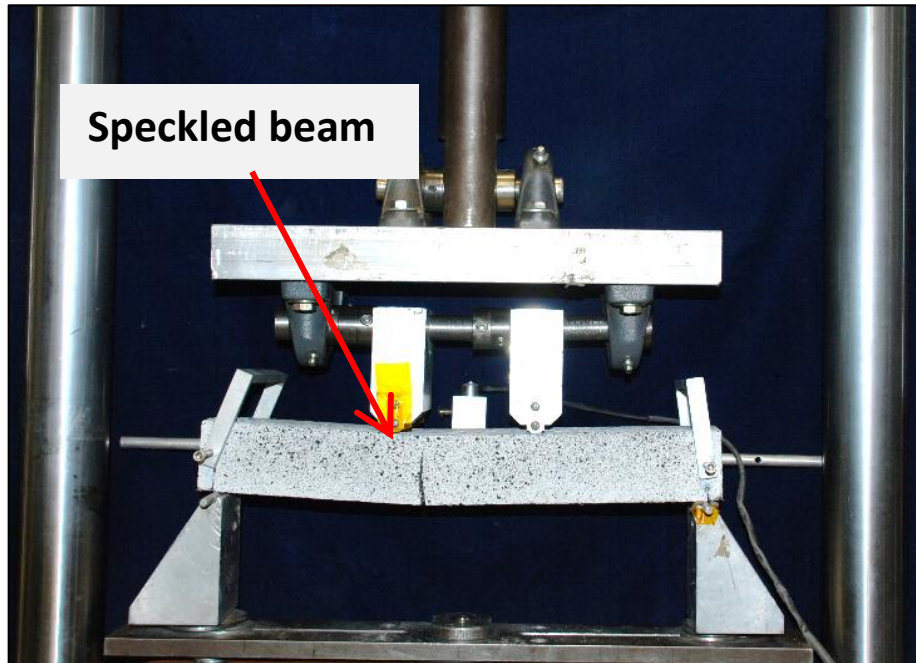


Flexural Closed Loop Testing

- 28 day flexural response was evaluated on prismatic specimens (2"x2.5"x15").
- Initial load-controlled mode was followed by displacement controlled mode @ 0.02 in/min until peak load and @0.004 in/min until failure.
- Midpoint deflection was measured using LVDT placed in the center of the bottom face of the beam
- Strain fields were monitored using digital image correlation (DIC) technique.

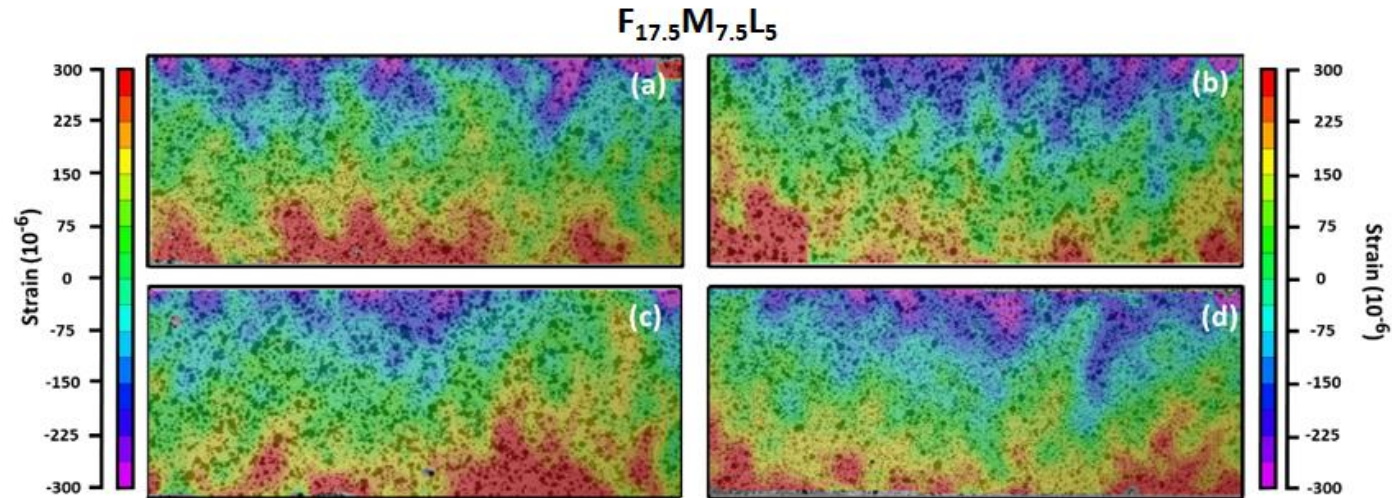


Flexural closed loop tests

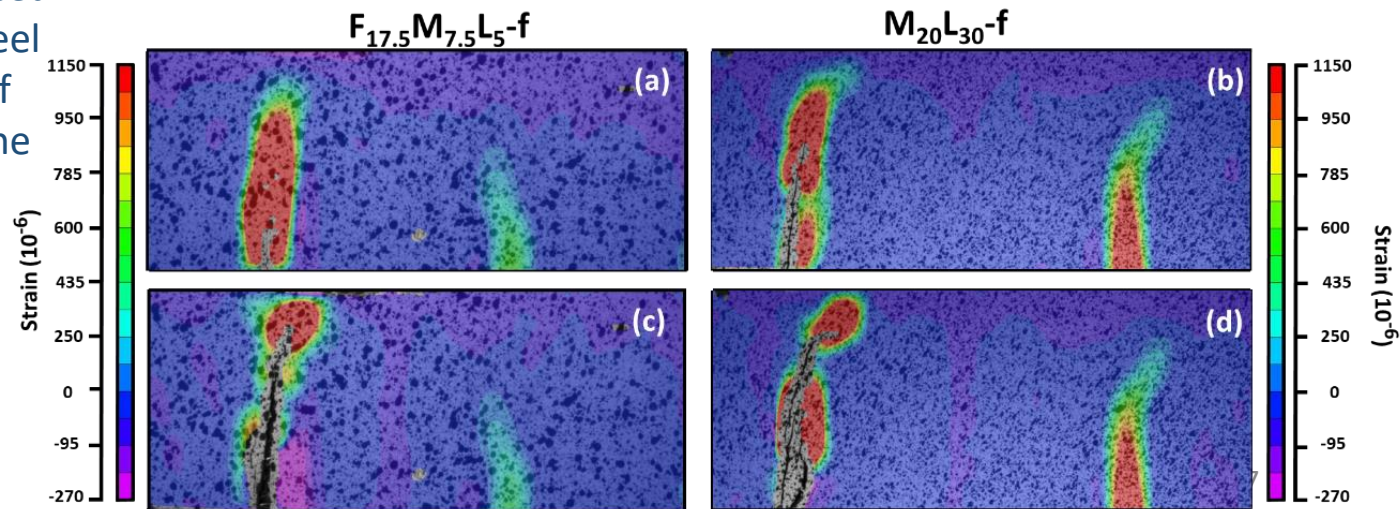


Flexural Test Results

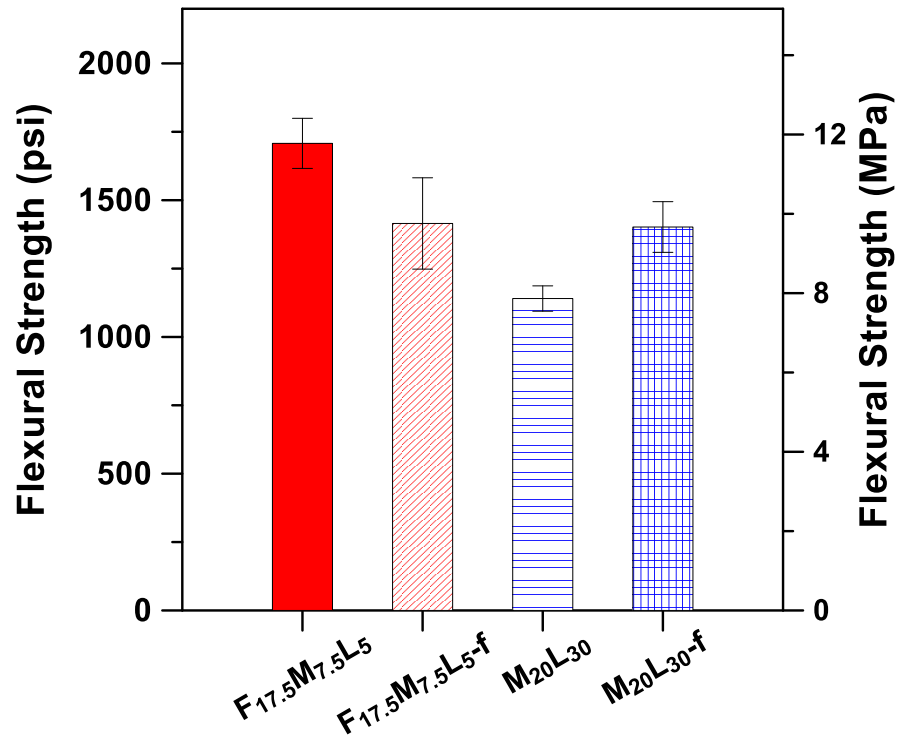
The strain distribution in four quaternary fly ash UHPC beams at peak load just prior to failure



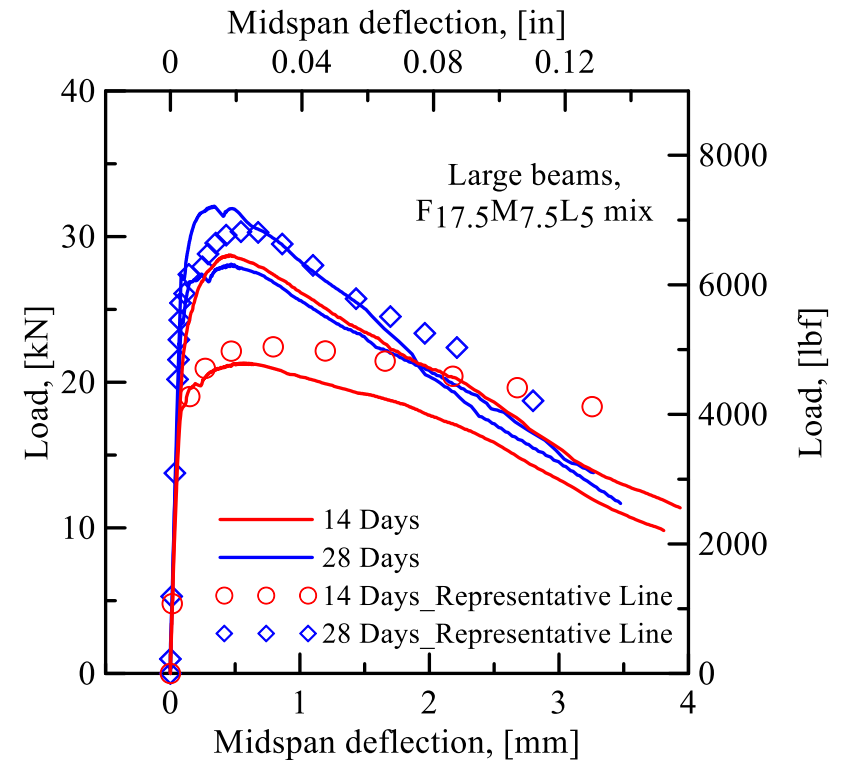
Strain fields for fiber reinforced beam specimens with 1% steel fiber) at a post peak stress of (a,b) 90% and (c,d) 50% of the peak stress.



Flexural Test Results



Load-Deflection response for non Fiber-reinforced mixes

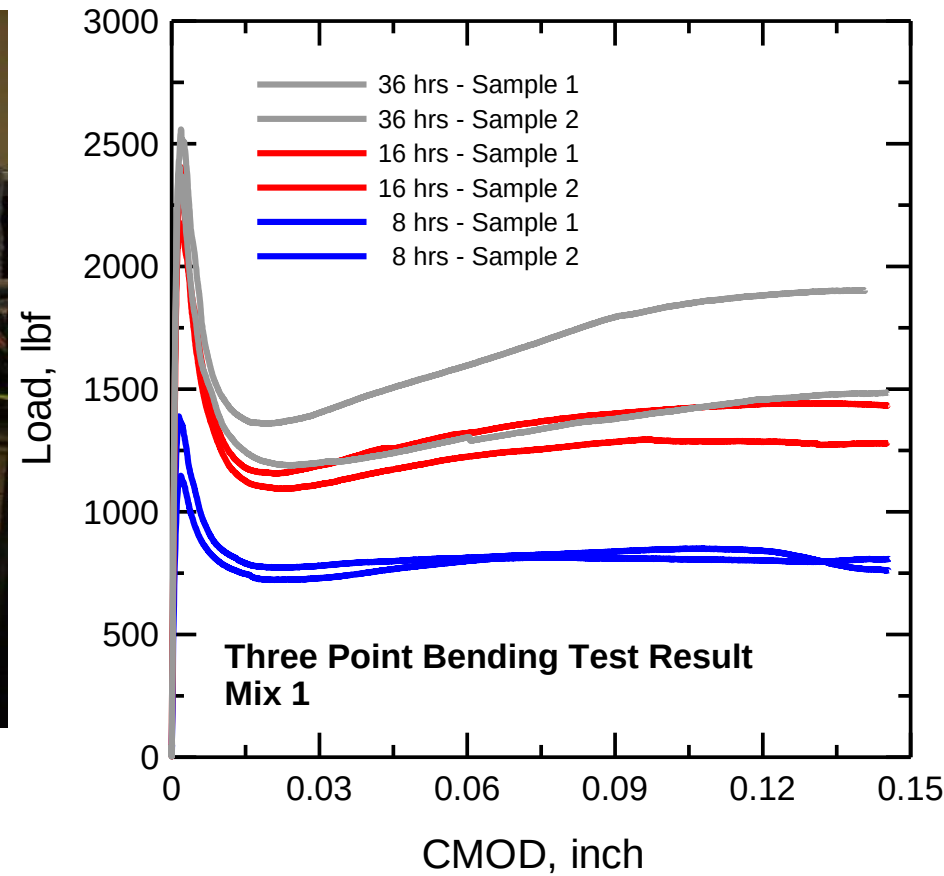


Load-CMOD response for Fiber-reinforced mixes

Design at the microstructure level will have a profound impact at the macro-structural level

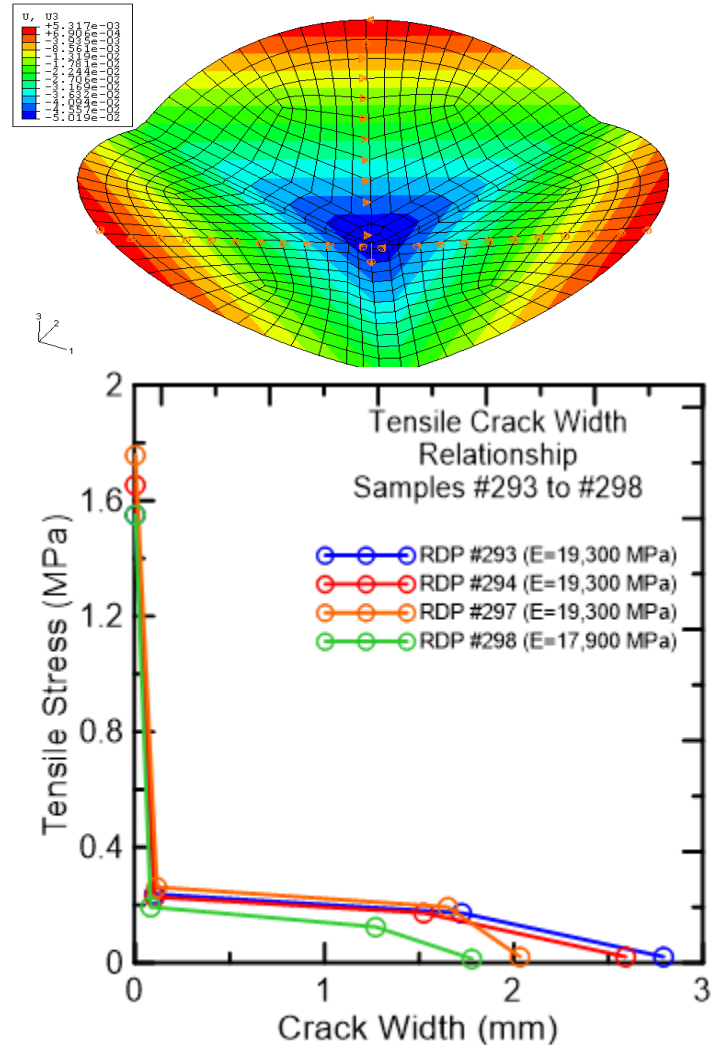
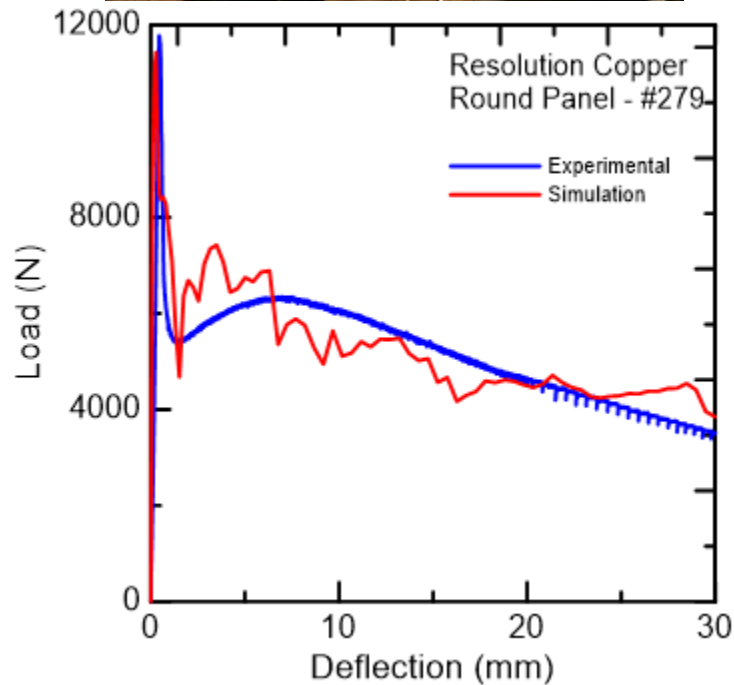
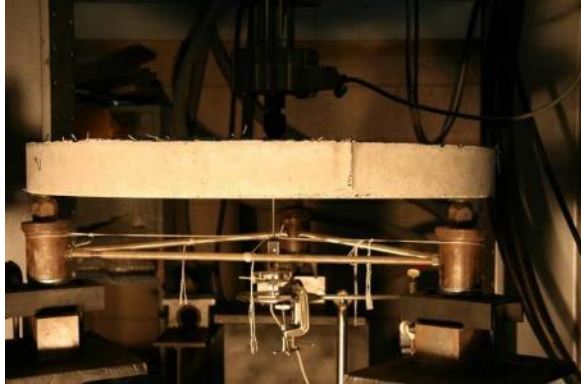


Consideration of Residual strength Effect



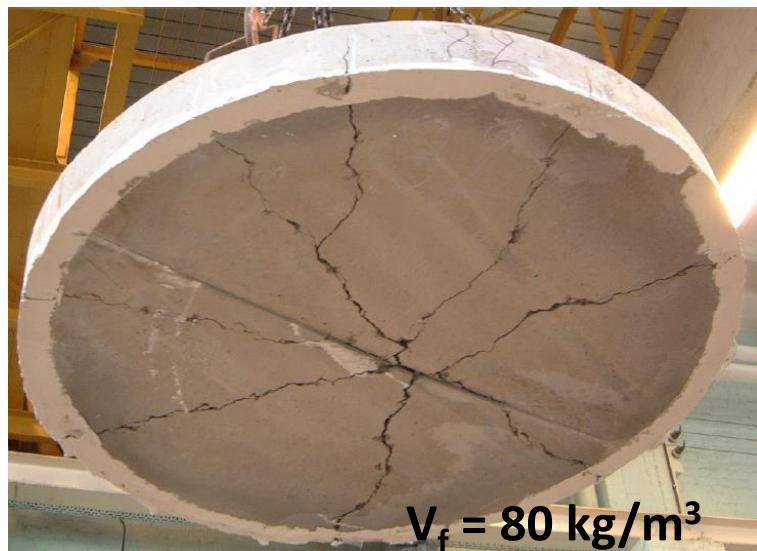
10-12 lbs/yd³ of macro fibers

ASTM C1550-Round Panel 3P-support specimen



Round panel tests for evaluation of SFRC

- Test setup
 - displacement controlled, continuous support, center point load
- Dimensions
 - $D, t = 1500, 150$ mm
 - stroke diameter = 150 mm

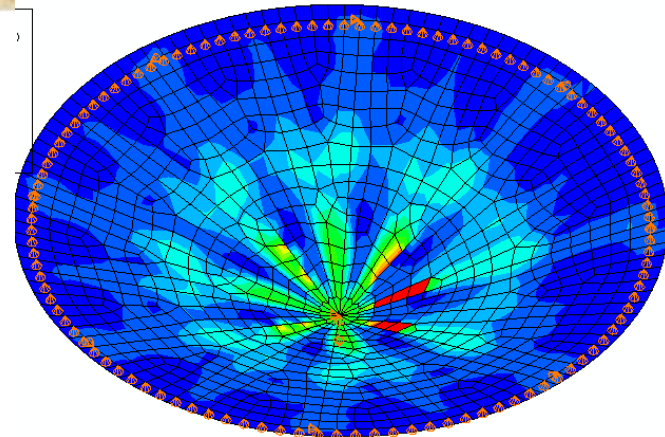
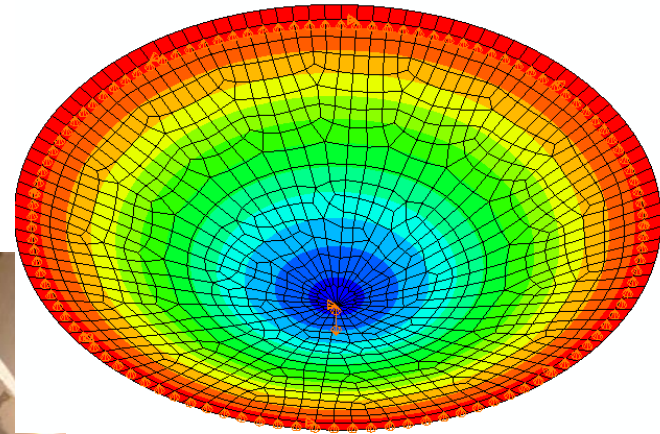
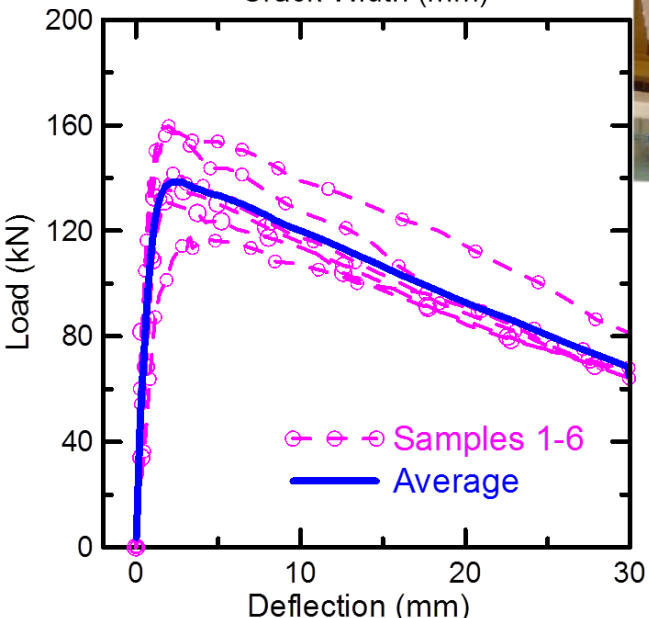
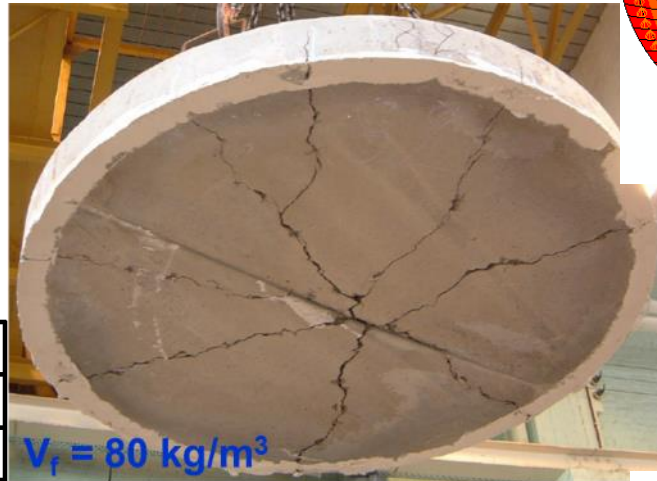
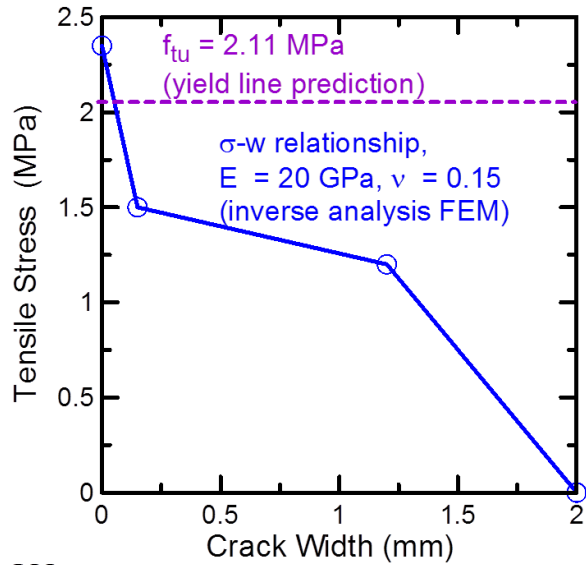


$V_f = 80 \text{ kg/m}^3$

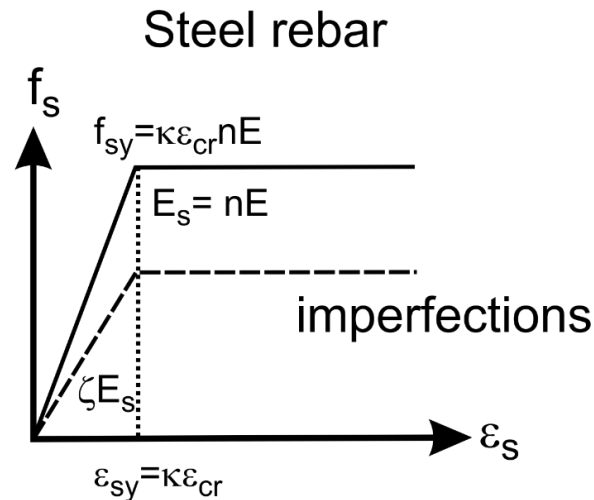
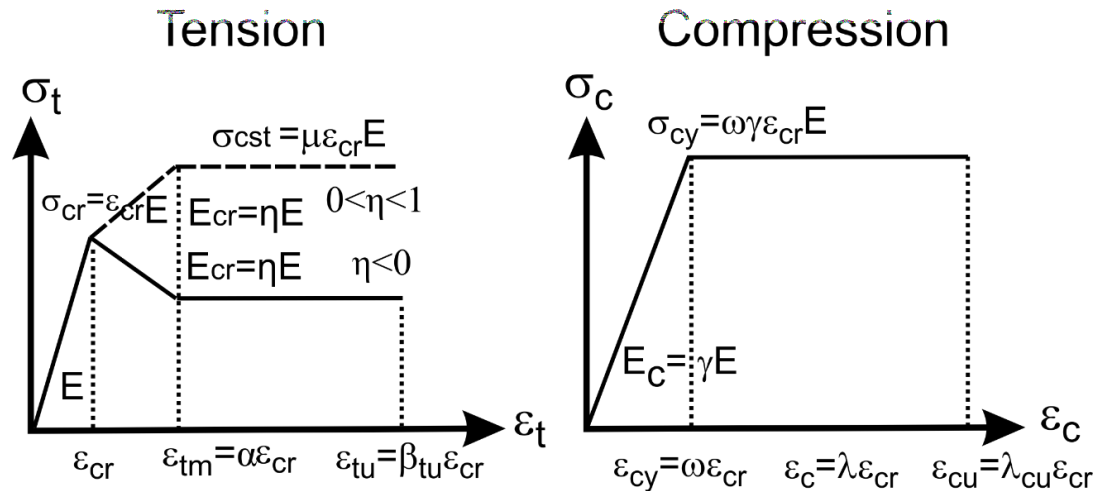


$V_f = 100 \text{ kg/m}^3$

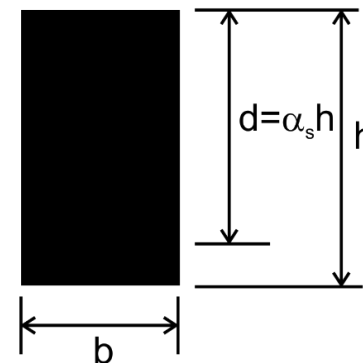
Round Panel Continuous Support Specimens



Material Models for Hybrid Design Approach



Cross section



Derivation of Moment-Curvature Relationship

Incrementally impose

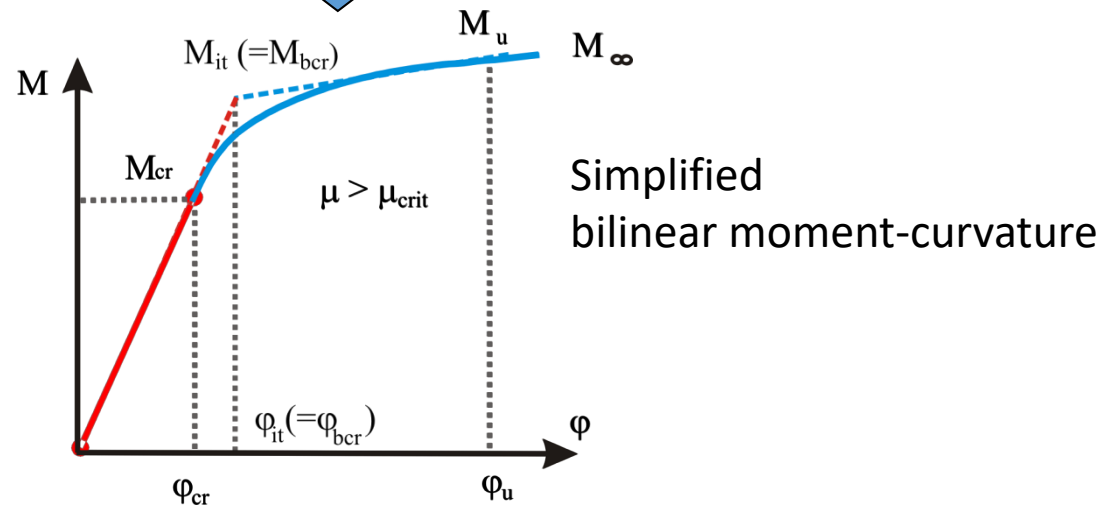
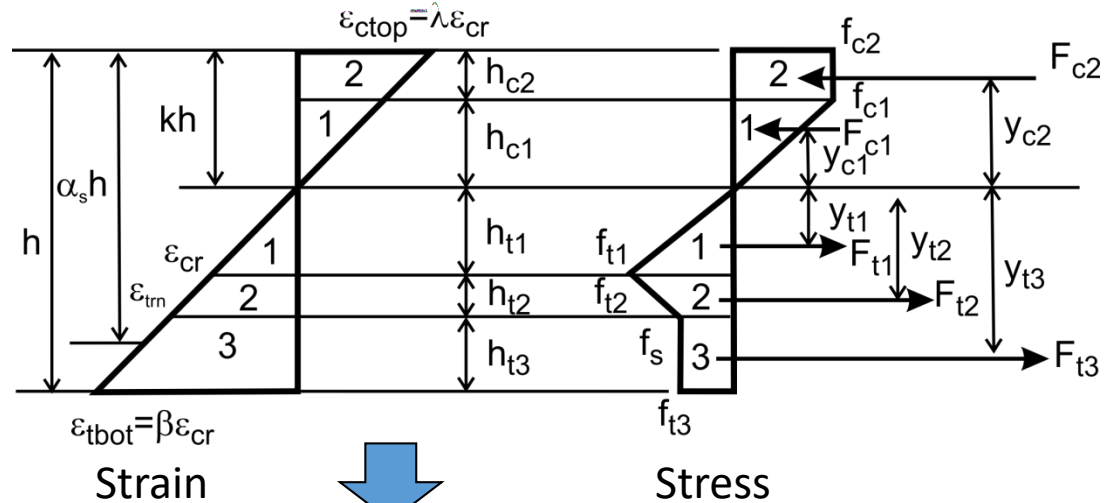
$$0 \leq \varepsilon_t \leq \varepsilon_{tu}$$

Strain Distribution
Stress Distribution

$\Sigma F = 0$, determine k

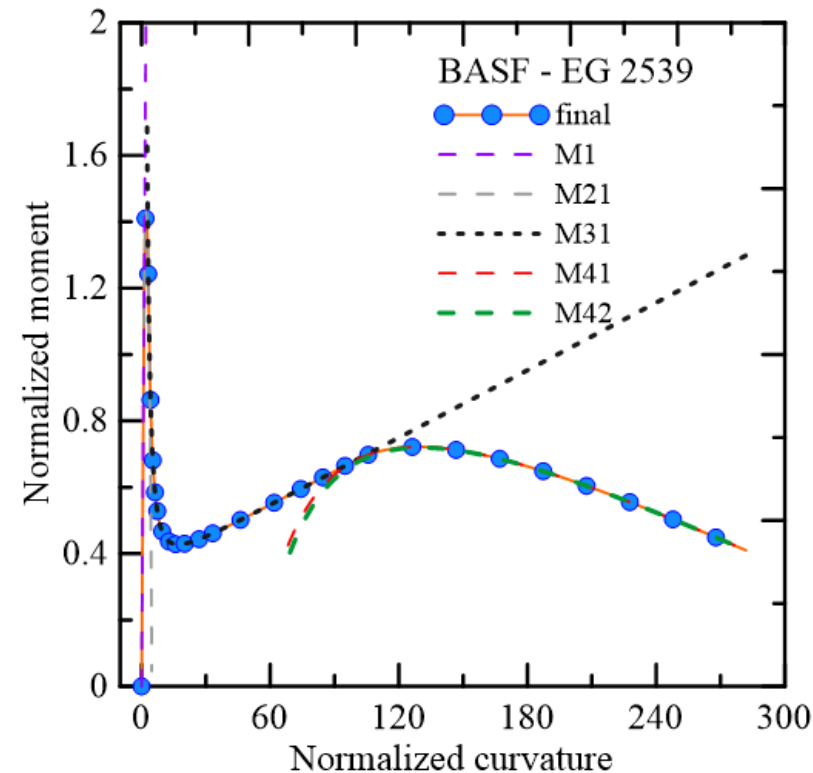
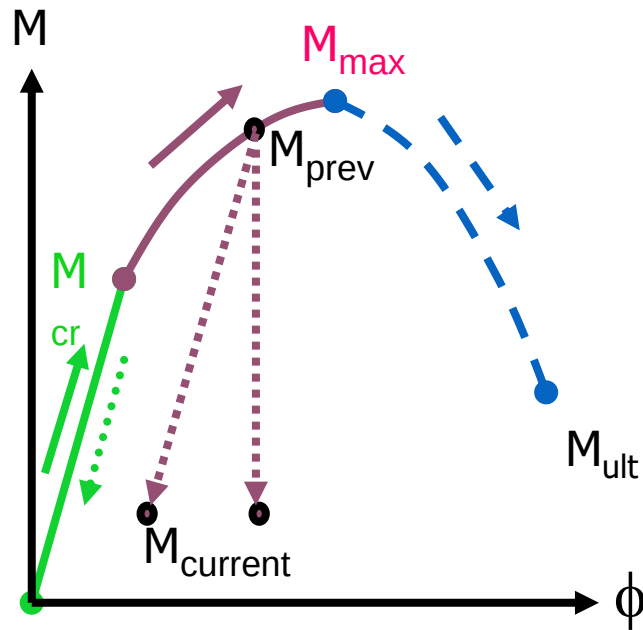
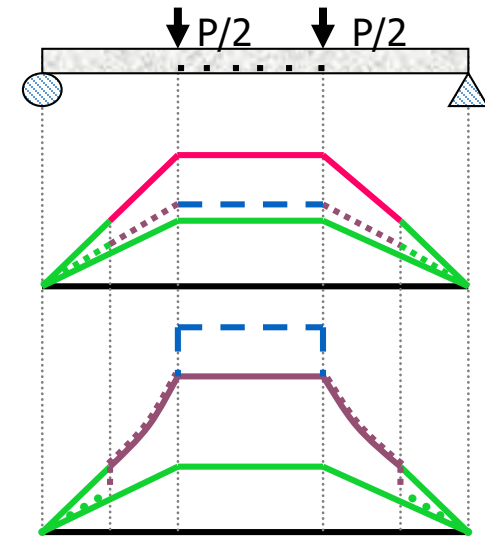
Moment: $M = \Sigma F_{ci} y_{ci} + \Sigma F_{ti} y_{ti}$
Curvature: $\phi = \varepsilon_c / kh$

Stage : $\lambda > \omega, \beta > \alpha$

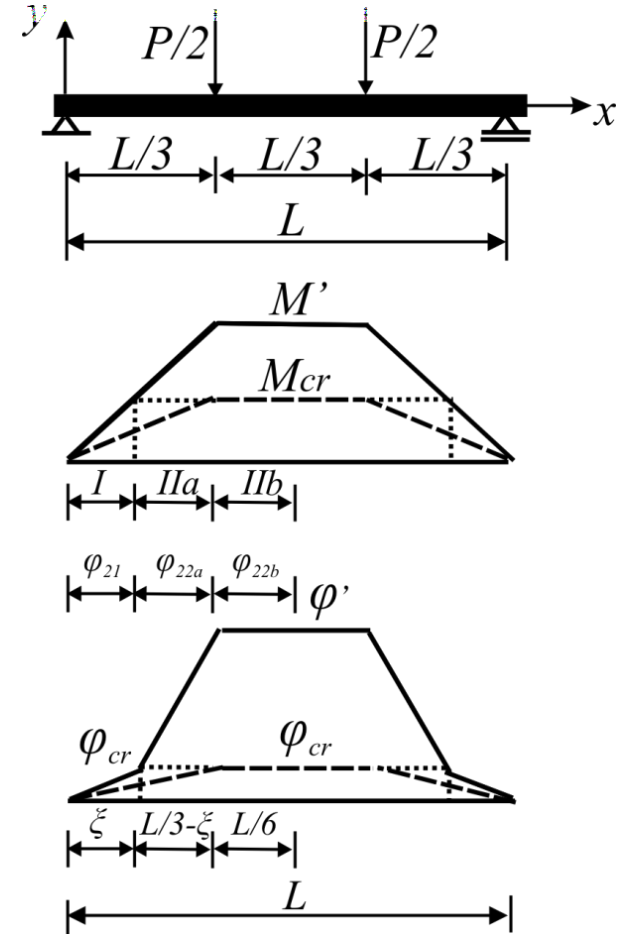
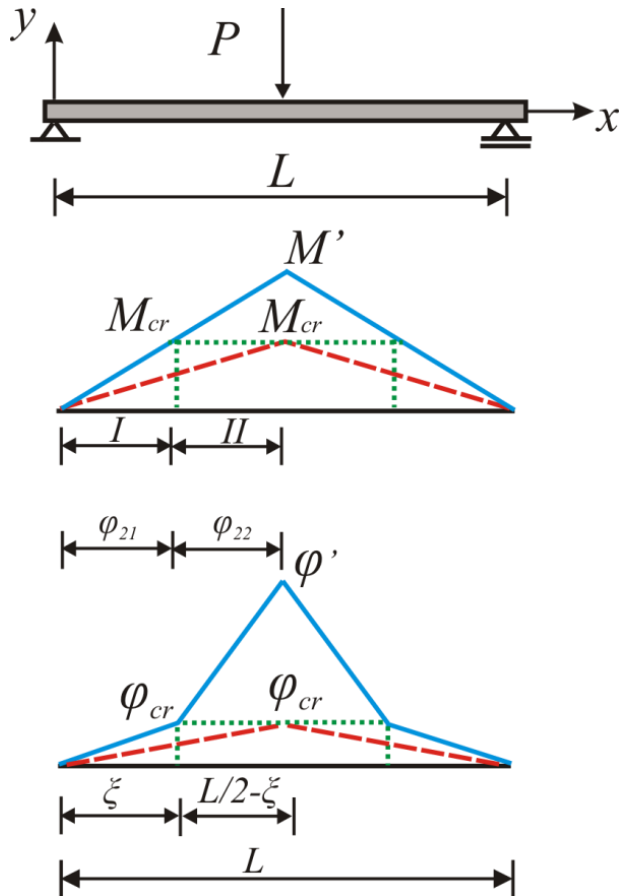


Load-Deflection Computation

- Moment-curvature diagrams are calculated for individual zones of response and then combined
- Equilibrium equations allow one to obtain support reactions and moment distribution
- The rotation and deflection distributions are obtained by the slope deflection equations



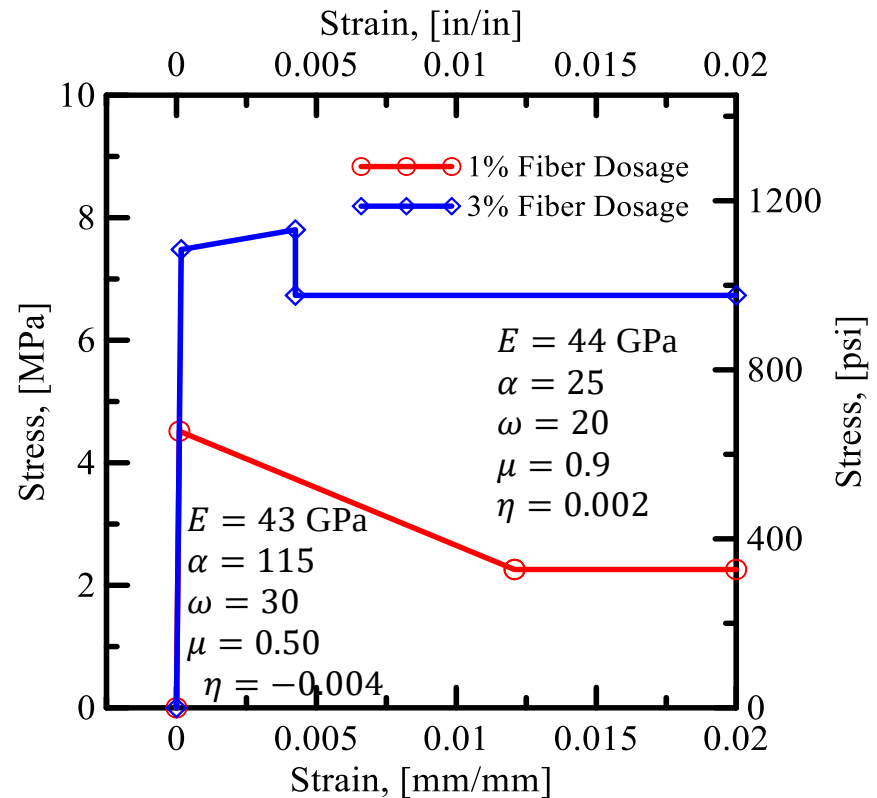
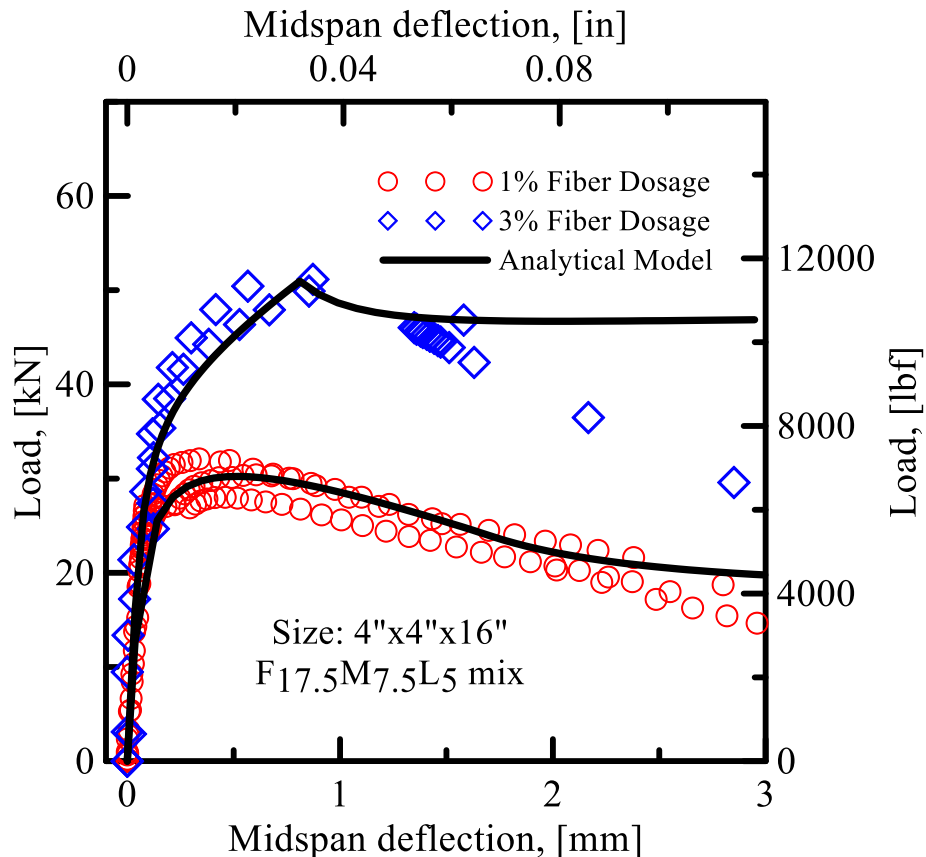
Moment-Curvature Distributions



- Use Static Equilibrium to get moment distribution
- Curvature distributions are generated in closed form using a bilinear moment-curvature model

Applications of the Constitutive Analytical Model

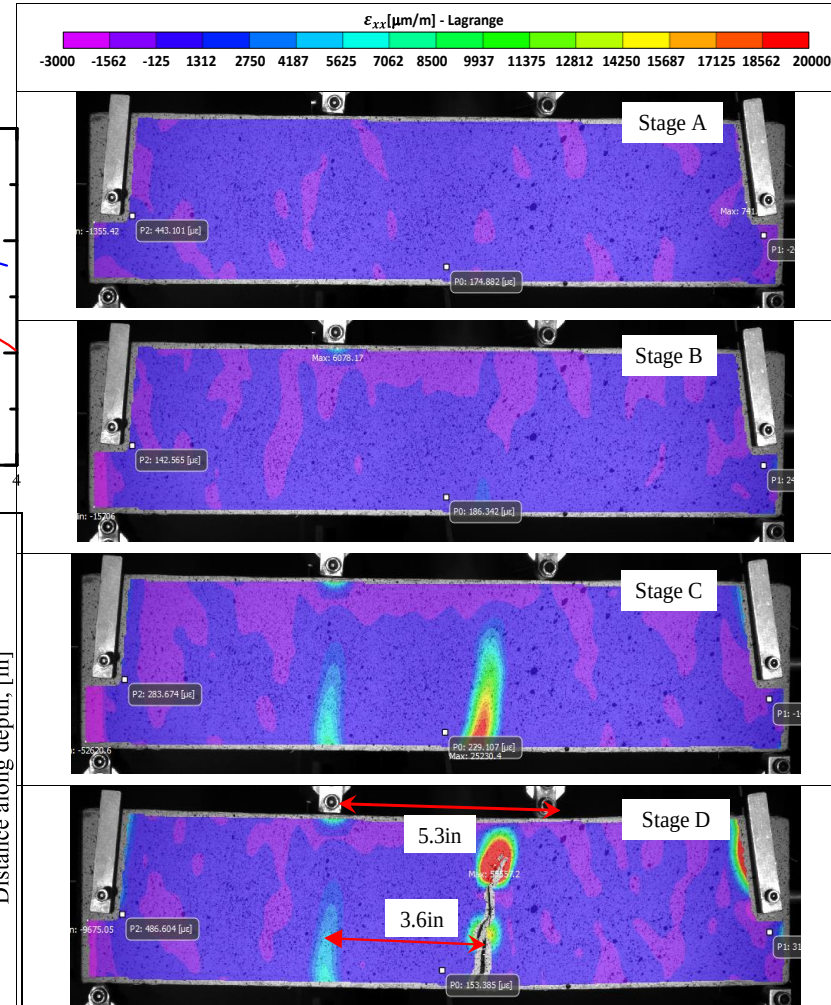
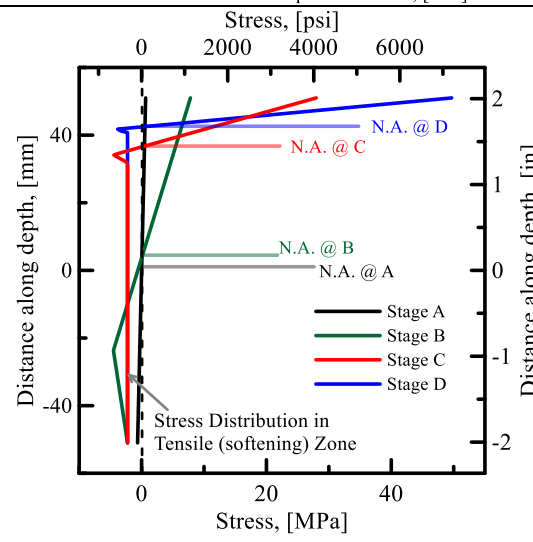
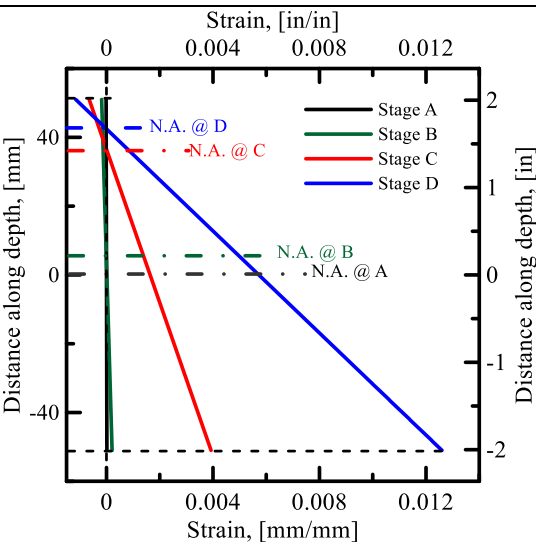
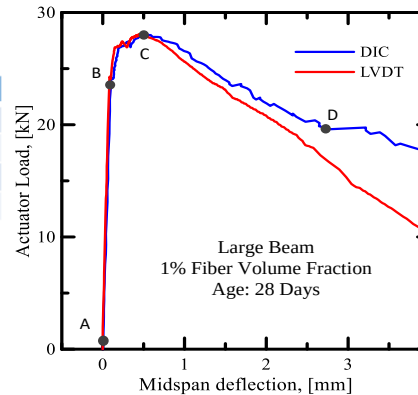
- Compute the Material Property Data Using Inverse Analysis



Applications of the Constitutive Analytical Model

Validation of the DIC Measurements in Flexural Crack Growth

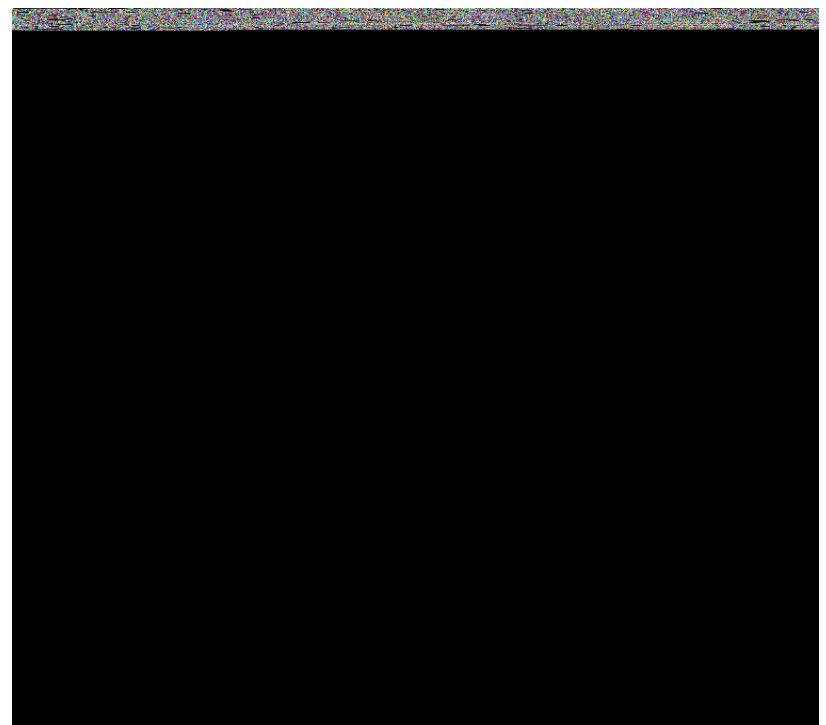
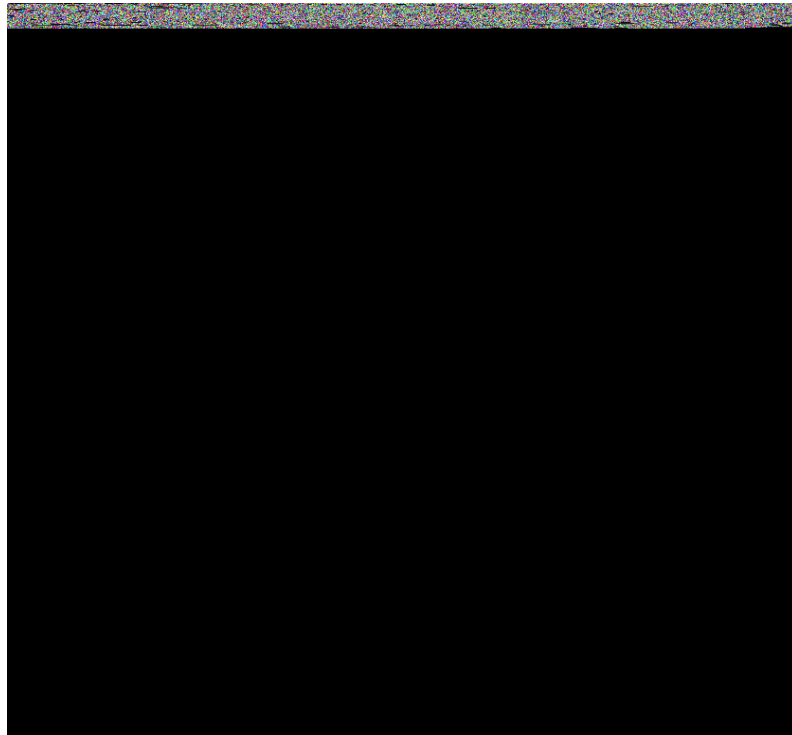
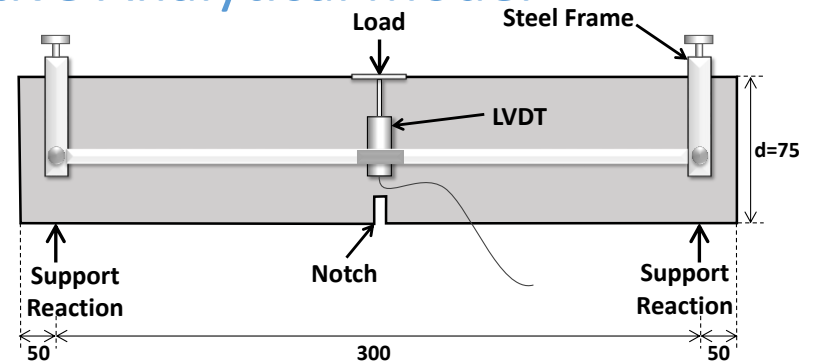
Stage	Characteristic
A	Initiation of the deformation
B	Initiation of the non-linear response
C	Response to the peak load
D	End of the DIC data analysis (beam failure)



Applications of the Constitutive Analytical Model

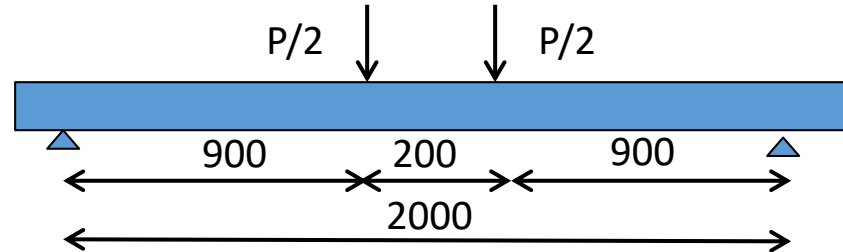
Comparison with other experimental results

Meng et al. UHPC beams with a dimension of 400x75x75 mm (16x3x3 inch).



Experimental Verification – Comparison with $\sigma-\varepsilon$ model

- Four-point bending test
- 200(b) × 200(d) × 2000 (L) mm
- Steel fibers: 50 kg/m³
- Rebar 2- ϕ 12



$\sigma-\varepsilon$ Model

$E = 29 \text{ GPa}$, $\varepsilon_{cr} = 130 \mu\text{str}$
 $\mu = 0.38$, $\beta_{tu} = 192$
 $\omega = 10$, $\lambda_{cu} = 27$, $\gamma = 0.7$

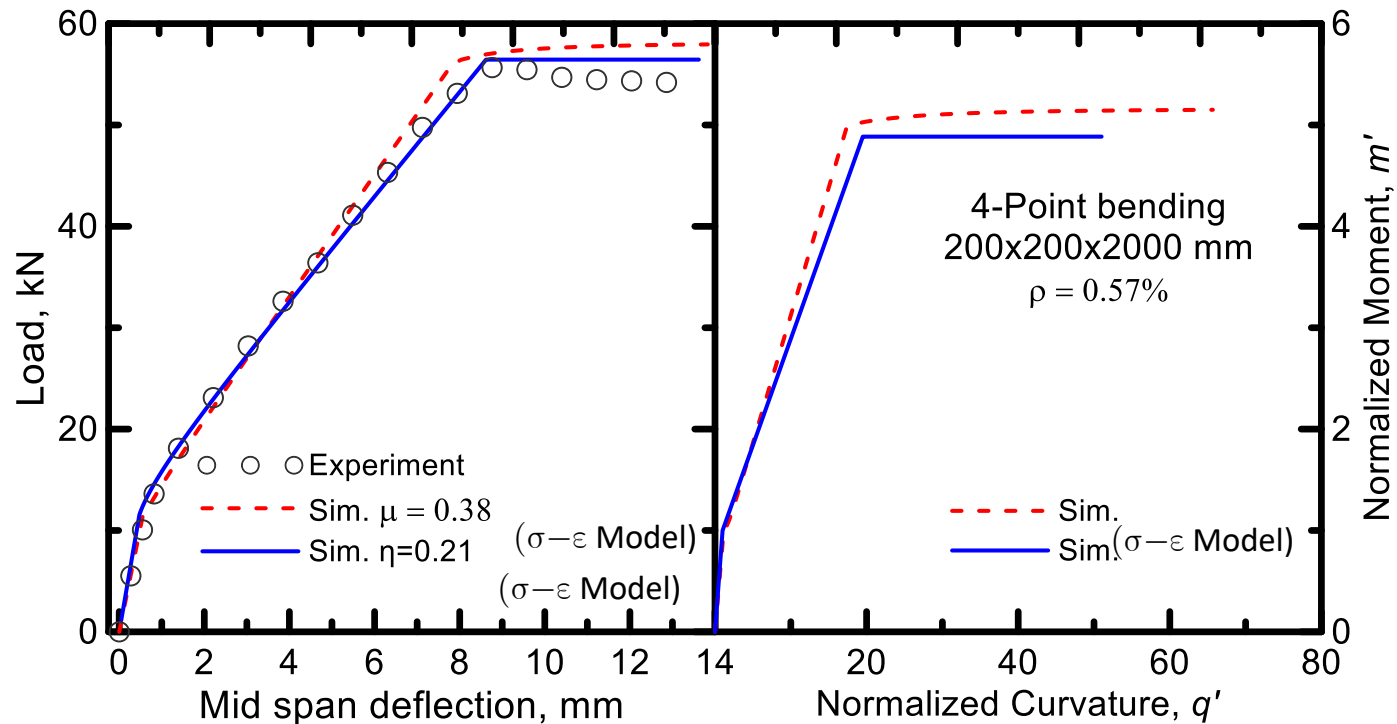
Bilinear M- ϕ Model

$E = 30 \text{ GPa}$, $\varepsilon_{cr} = 130 \mu\text{str}$
 $\eta = 0.21$, $m = 4.88$, $q = 20$



$$EI_g = 4.0 \times 10^{12}$$

$$EI_{cr} = 8.4 \times 10^{11}$$



Dupont D. "Modelling and experimental validation of the constitutive law ($\sigma-\varepsilon$) and cracking behavior of steel fiber reinforced concrete" Ph.D. Dissertation, Catholic University of Leuven, Belgium, 2003

Experimental Verification – RC with Steel fibers

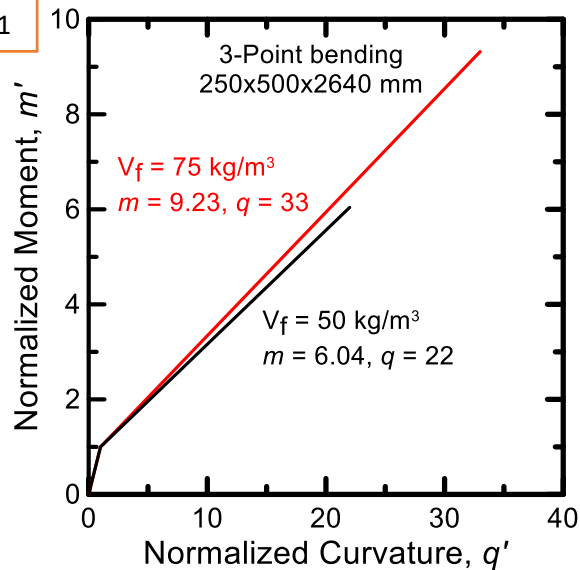
- Three-point bending test
- Fiber content: 50 and 75 kg/m³

Beams	Group #1	Group #2	Group #3
Height (mm)	500	1000	1500
Span(mm)	2640	5640	8640
Width(mm)	250	250	250
Reinforcement Longitudinal bars	8-φ14	8-φ20	8-φ24

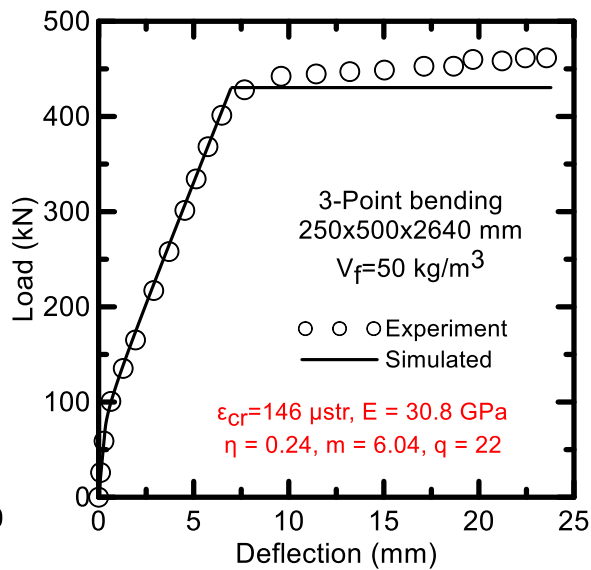
Simulated Parameters

Beam Type	Fiber Content kg/m ³	ϵ_{cr} 10 ⁻⁶	E GPa	η	m	q	EI_g , 10 ¹³	EI_{cr} , 10 ¹³
Group #1	50	146	30.8	0.24	6.04	22	8.02	1.93
	75	120	31	0.26	9.23	33	8.07	2.10
Group #2	50	120	30	0.22	5.18	20	62.5	13.8
	75	120	31	0.28	6.88	22	64.6	18.1
Group #3	50	120	29	0.24	6.04	22	204	48.9
	75	120	30	0.28	7.16	23	211	59.1

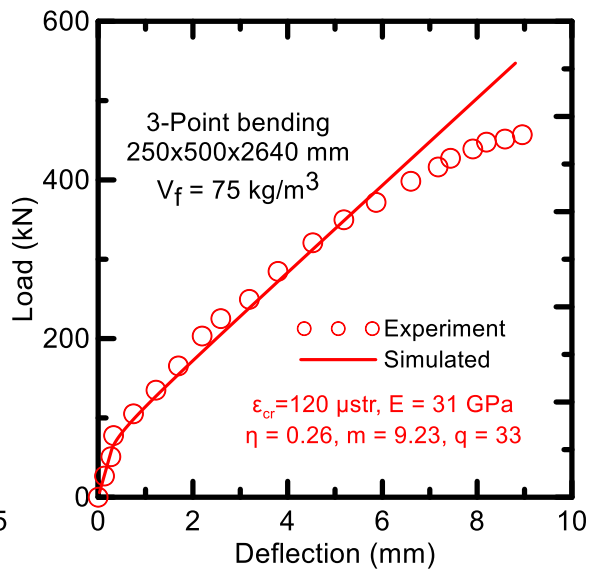
Group #1



Normalized curvature vs. Normalized moment

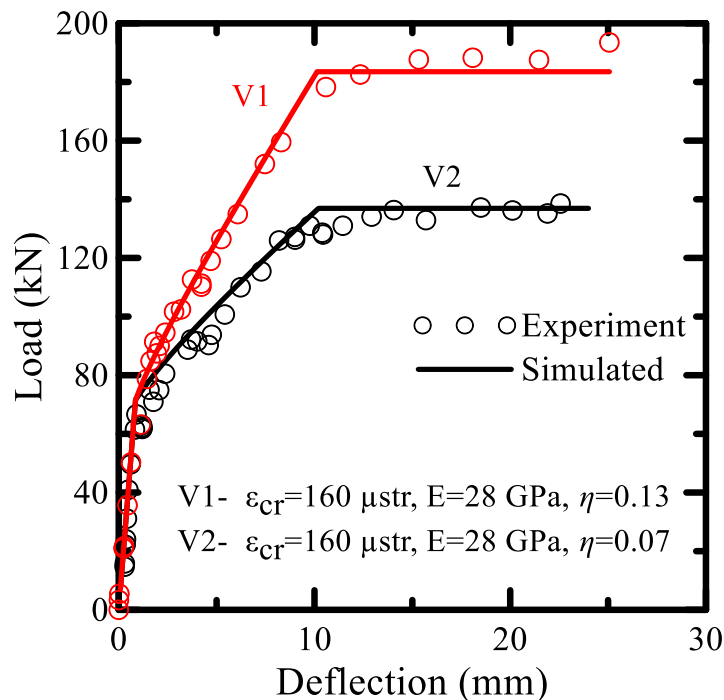


Deflection at mid-span vs. Applied load



Experimental Verification – RC with Steel fibers

- Four-point bending test
- 200(b) × 600(d) × 4000 (L) mm
- Steel fibers
- Longitudinal and shear reinforcement
- Measure deflections by five LVDTs
- $x = 750, 1500, 2000, 2500, 3250$ mm



Deflection at mid-span vs. Applied load



Details of beams

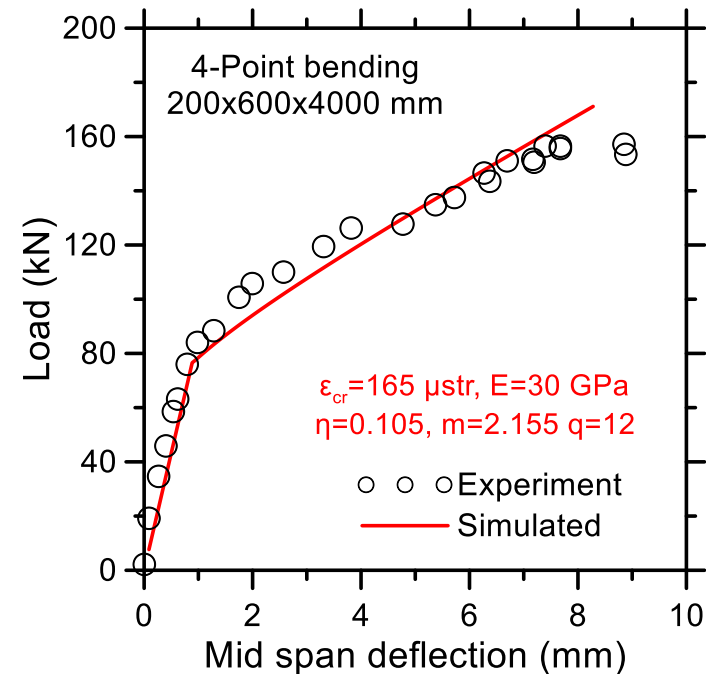
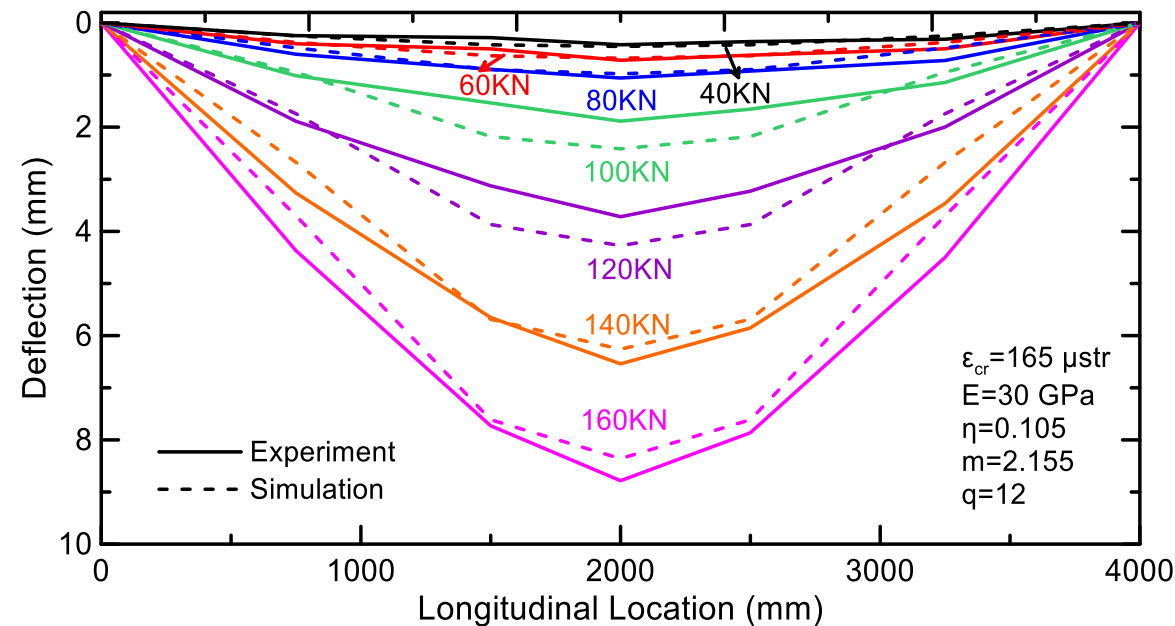
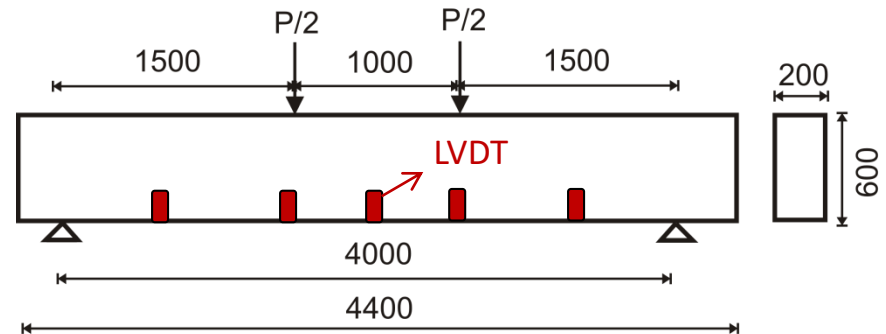
Beam ID	Fiber volume fraction (%)	Longitudinal reinforcement			Shear reinforcement	
		Ratio (%)	Tension	Top bar	Ratio (%)	Rebars
V1	0	0.262	4 ϕ 10	2 ϕ 6.3	0.104	4 ϕ 6.3
V2	0	0.262	4 ϕ 10	2 ϕ 6.3	0	0

Simulated Parameters

ID	m	q	$EI_g, 10^{14}$	$EI_{cr}, 10^{14}$
V1	2.56	100	1.01	0.131
V2	1.91	92	1.01	0.071

Experimental Verification– RC with Steel fibers

- 4m span
- Fiber volume fraction : 1.25%
- 2 ϕ 10 (Tension bars), 2 ϕ 6.3(Comp. bars)
- Pre-cracked stiffness $EI_g=1.08 \times 10^{14}$
- Post-cracked stiffness $EI_{cr}=1.13 \times 10^{13}$

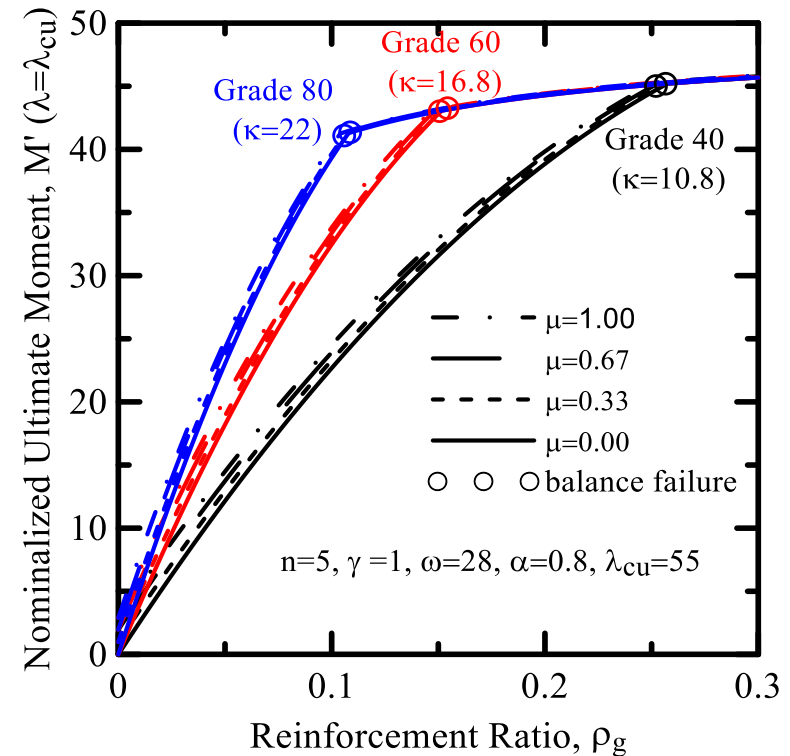
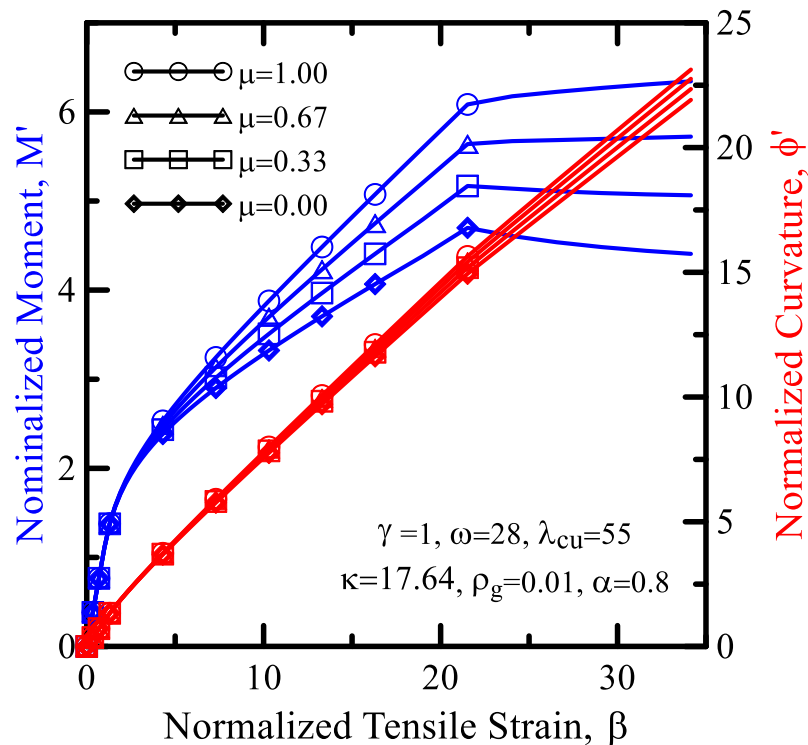


Hybrid Model for combination of Reinforcement and Fibers

For Hybrid Reinforced Concrete (HRC) Beams

$$M_n \approx m_{\infty} M_{cr} = \frac{6\rho_g n\kappa(\mu\alpha - \mu + \alpha\omega) + 3\omega\mu - 3(\rho_g n\kappa)^2}{\omega + \mu} M_{cr}$$

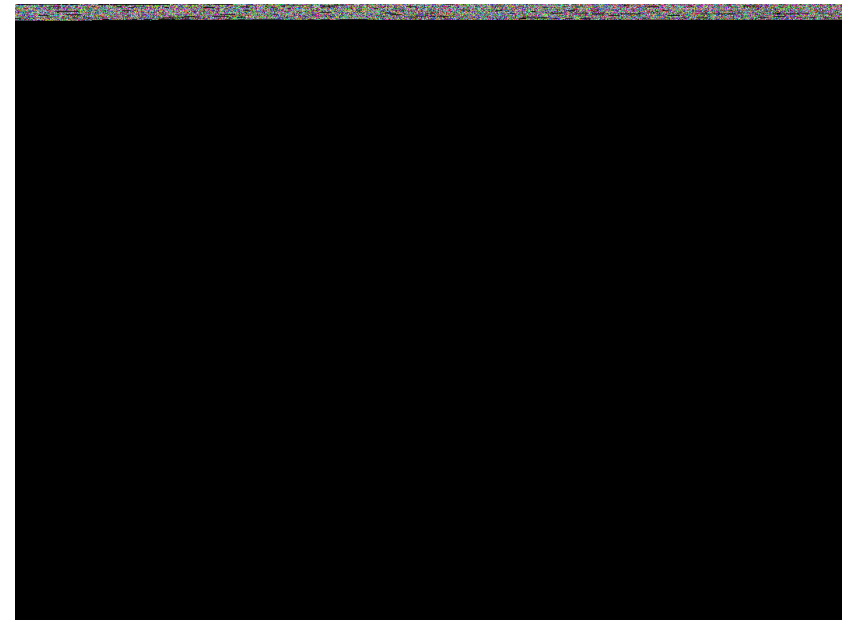
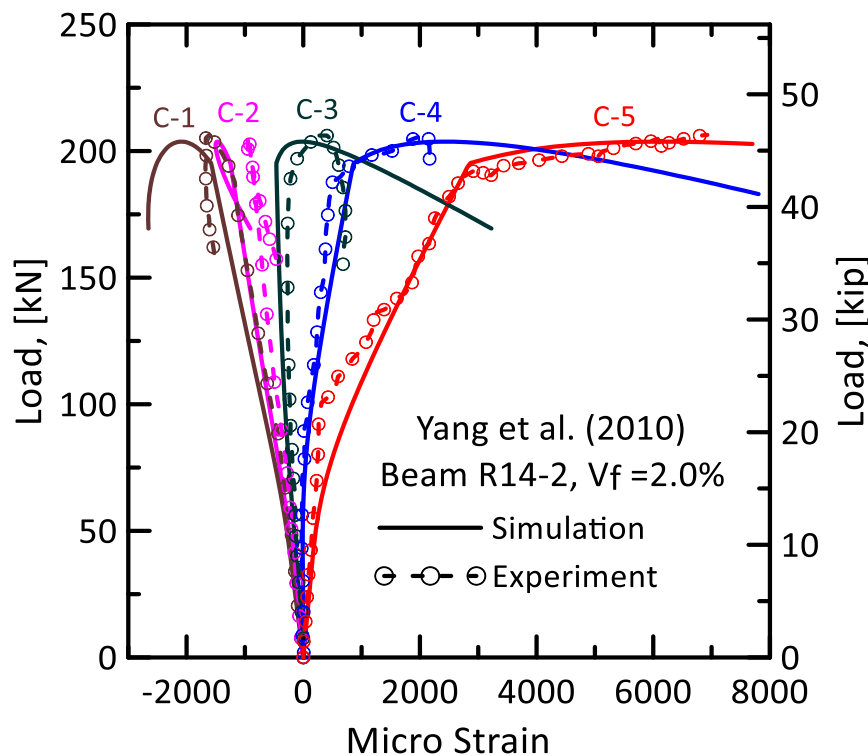
$$\omega = \frac{\varepsilon_{cy}}{\varepsilon_{cr}} \approx \frac{f'_c}{\sigma_{cr}} = \frac{0.85 f'_c}{0.04 f'_c} = 21.25$$



Applications of the Constitutive Analytical Model

- Comparison with other experimental results on HRC

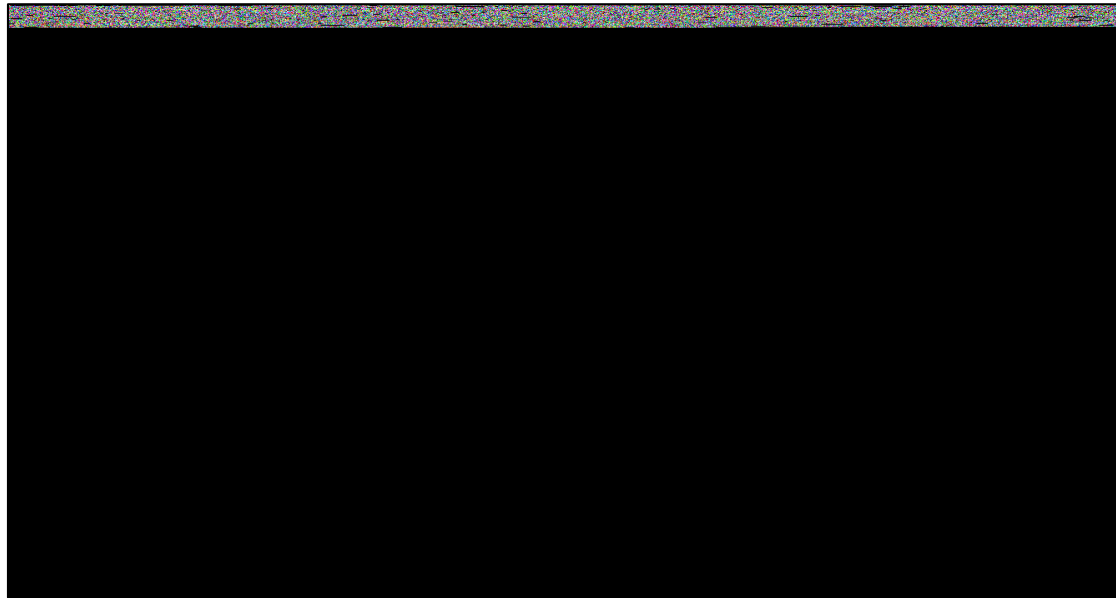
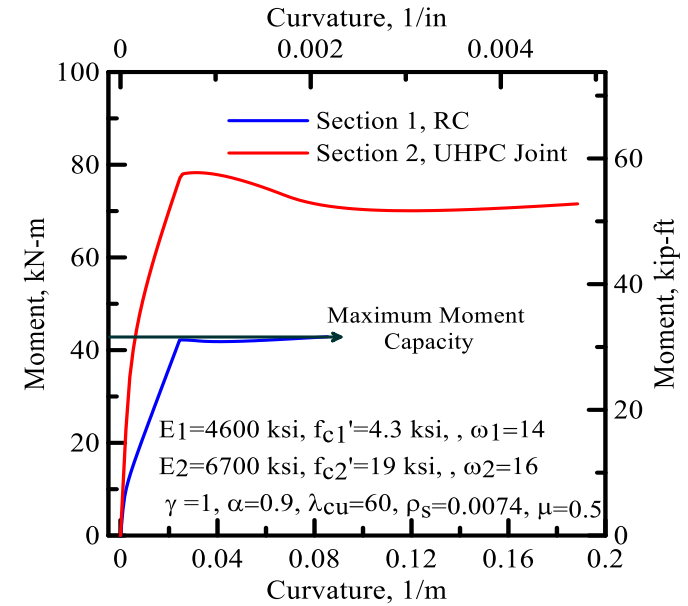
Yang et al. (2010), experimental tests on flexural behavior of ultra-high performance concrete (UHPC) beams. Strain distributions at different points along the beam depth



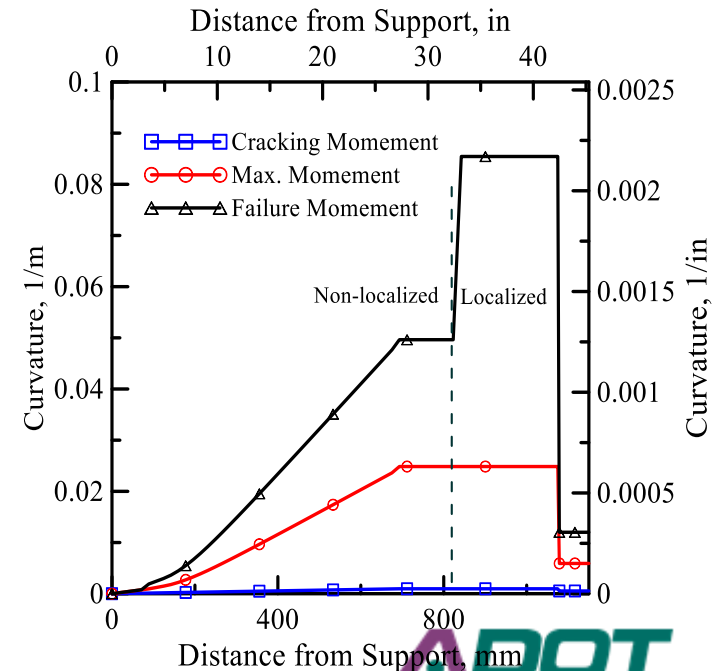
predicting the strains distributions along the beam depth

Evaluation of UHPC Connection Pours

- Two sections modelled, regular RC and UHPC
- Each section with its own moment-curvature response
- The moment capacity is dominated by the capacity of the weaker section (RC section)
- These curves show that making stronger joints will not help to improve the capacity of the member or slab



Haber et al., executed an experimental study on six different commercially-available materials (UHPC type A to F)



Conclusions

- Selected concrete mixture combinations including the use of 2 volume fractions of fibers, with a detailed cost analysis showing that these combinations cost 40-60% less than proprietary mixtures per unit strength.
- Work with the fibers to less than 1.5% volume fraction.
- Evaluated compressive strength development, flexural strength development, fracture properties, shrinkage cracking of the selected mixtures. Compare them with a traditional HPC.
- Analytical closed form equations to predict the load-deflection response of UHPC based on bilinear moment-curvature relationship
- The methods apply to all type of flexural members exhibiting deflection-softening and hardening behavior
- Accuracy of the proposed model was identified by comparing the simulated data with experimental data

Acknowledgements

- This project was partially supported by the Arizona Department of Transportation ADOT, SPR 745.
- Support of ADOT Research Center and Bridge Engineering group is greatly appreciated.